Lecture 12: Data Visualization

CMSC/DATA 320: Introduction to Data Science

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Announcements

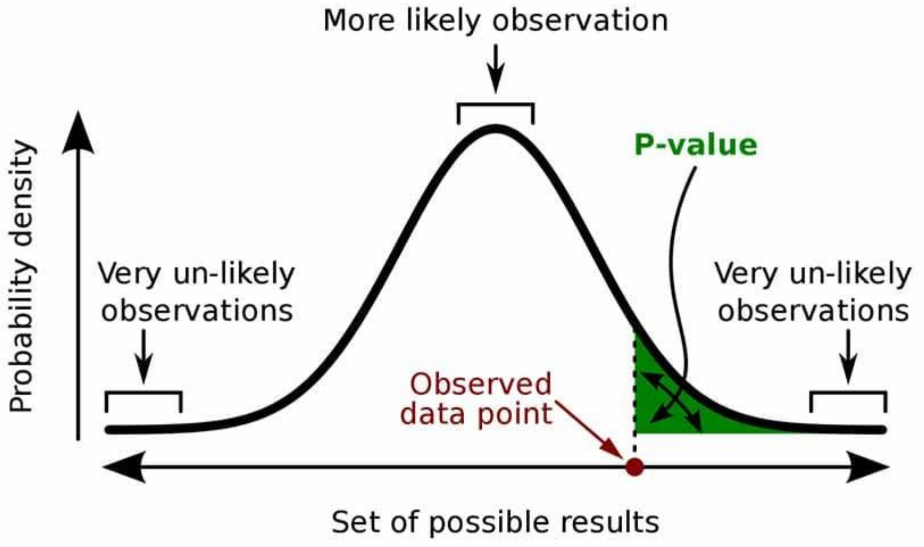
- PA3, WA3 due Sunday
- Mid-semester check-in assignment available soon
  - Also asks about office hour preferences

finishing up Hypothesis Testing

p-value of a statistical test

A measure used to **quantify the strength of evidence against the null hypothesis**.

- ❑ Represents the probability of observing the test statistic, or a more extreme value, *if the null hypothesis is true*.
  - ○ p-value = P(statistic | H_0 = true)
- ❑ A low p-value indicates that the observed data are unlikely under the null hypothesis, suggesting strong evidence against it.
- ❑ If the p-value ≤ α, the null hypothesis is rejected.



A **p-value** (shaded green area) is the probability of an observed (or more extreme) result assuming that the null hypothesis is true.

Calculating p-value

As with test statistic, calculation of p-value **depends on the test you are applying**.

In python, typically `scipy.stats` tests will return *both* the test statistic and the p-value when you call the corresponding statistical test function.

Example 1: Calculating test statistic and p-value

2 sample t-test: test the means of 2 independent samples

Consider a dataset containing education levels. We want to compare average education level between people with assigned sex male vs. female.

H_0: $\mu_{male} = \mu_{female}$

H_a: $\mu_{male} \neq \mu_{female}$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

```
from scipy.stats import ttest_ind

t_statistic, p_value = ttest_ind(females, males)
```

output:

```
t-statistic: 1.1280639480025711
p-value: 0.2592940410499226
```

`ttest_ind(GROUP1_DATA,GROUP2_DATA)`

Example 2: Calculating test statistic and p-value

1 sample t-test: test the mean of 1 sample against the population mean

Consider a dataset containing monthly births for a given year. The population mean has been considered to be 10,500 births/month. A researcher believes this is no longer the case.

H_0: μ = 10,500

H_a: μ ≠ 10,500

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

```python
from scipy.stats import ttest_1samp

result = ttest_1samp(df['births'], popmean=10500.00)

print("t-statistic:", result.statistic)
print("p-value:", result.pvalue)
```

output:

```
t-statistic: 3.6798245086768624
p-value: 0.000268664803791436
```

```
ttest_1samp(YOUR_DATA, popmean=POPULATION_MEAN)
```

Compare a p-Value to α

Statistical Significance

Def: An outcome is said to be statistically significant if it is **unlikely to have occurred due to chance**.

Recall: H_0 is the statement being tested. It proposes that no statistical significance exists for a given set of observations.

Statistical Significance

Def: An outcome is said to be statistically significant if it is **unlikely to have occurred due to chance**.

Recall: H_0 is the statement being tested. It proposes that no statistical significance exists for a given set of observations.

What do we mean by "unlikely to have occurred due to chance"?

Statistical Significance and p-value

If the p-value we calculated is <u>lower</u> than α, the outcome is **statistically significant**, and thus the null hypothesis is rejected. Specifically:

- p-value $\leq \alpha$: reject H_0
- p-value $> \alpha$: fail to reject H_0

If the outcome is statistically significant, we are confident that it is real, rather than due chance of the chosen the sample.

Example: Chosen $\alpha = 0.05$. Calculated p-value is 0.13.

$\rightarrow$ p-value $> \alpha$

$\rightarrow$ Null hypothesis is NOT rejected

$\rightarrow$ Outcome is NOT statistically significant

Compare p-value with α

Example: **1 sample t-test**: test the mean of 1 sample against the population mean

Consider a dataset containing monthly births for a given year. The population mean has been considered to be 10,500 births/month. A researcher believes this is no longer the case.

H_0: μ = 10,500

H_a: μ ≠ 10,500

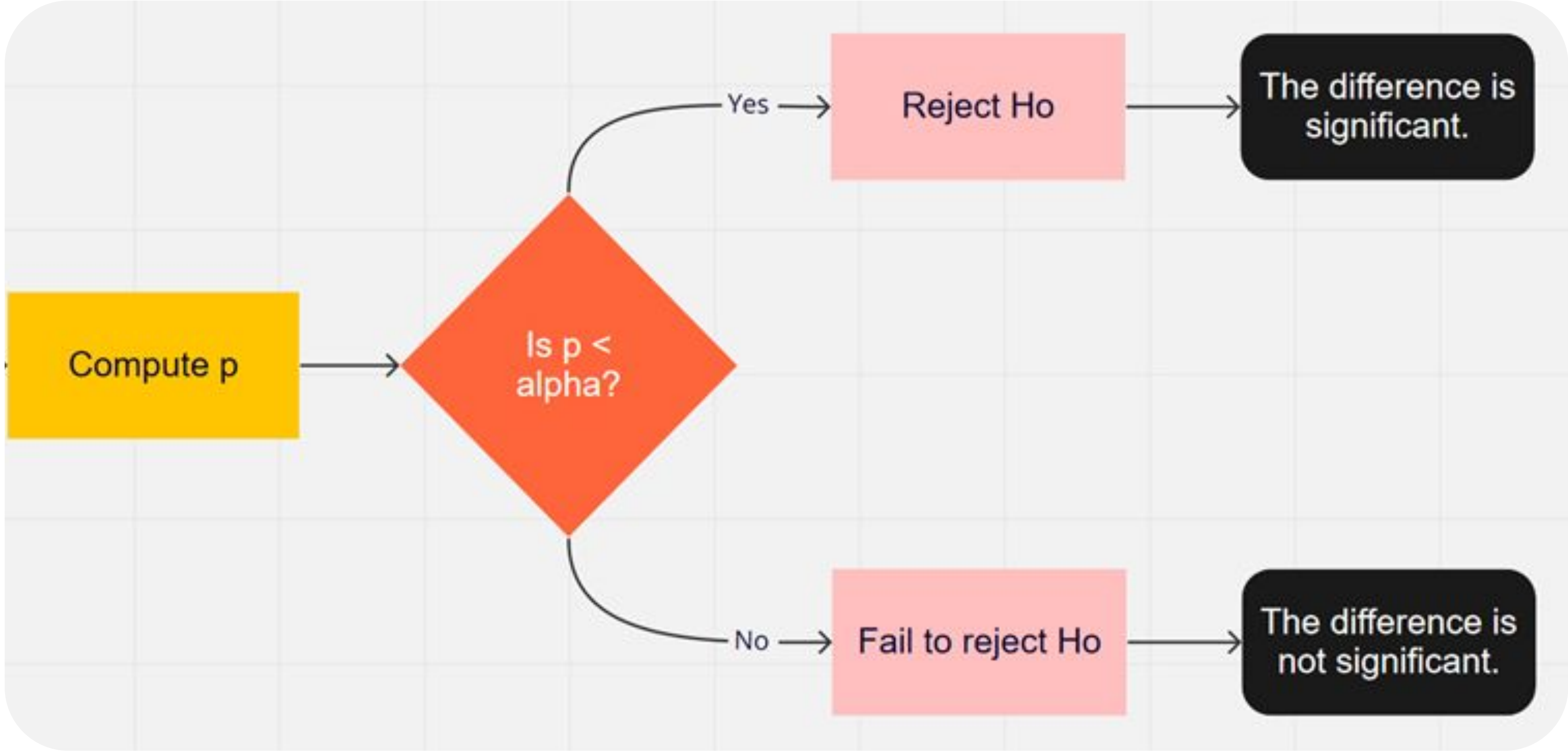
output:

```
t-statistic: 3.6798245086768624
p-value: 0.0026864803791436
```

Suppose we had chosen α = 0.01.

→ What is our required confidence level?
→ Is this outcome statistically significant?
→ Do we reject H_0 (that the population average is 10,500)?

Recap



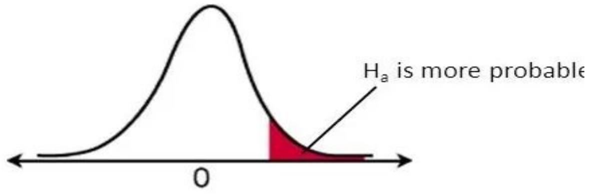
Compute p
Is p < alpha?
Yes
Reject Ho
The difference is significant.
No
Fail to reject Ho
The difference is not significant.

One-tailed vs Two-tailed tests

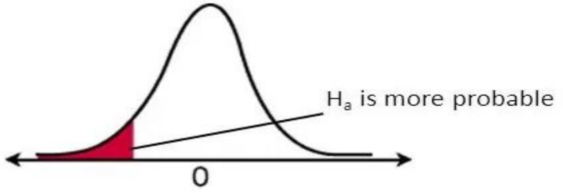
We can talk about the critical region by describing which "tails" of the probability curve we are concerned with.

- **One-tailed** = looking at the tail of the curve on *either* the left or the right
  - Usually specified as "left-tailed" or "right-tailed"
- **Two-tailed** = looking at tails on *both* the left and the right

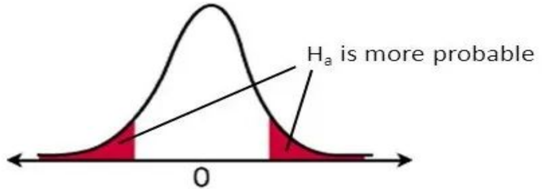
In this way we specify the **directionality of the test** and the **critical region**, and thus how we will determine which values of the test statistic lead to the **rejection of the null hypothesis**.



Right-tailed test



Left-tailed test



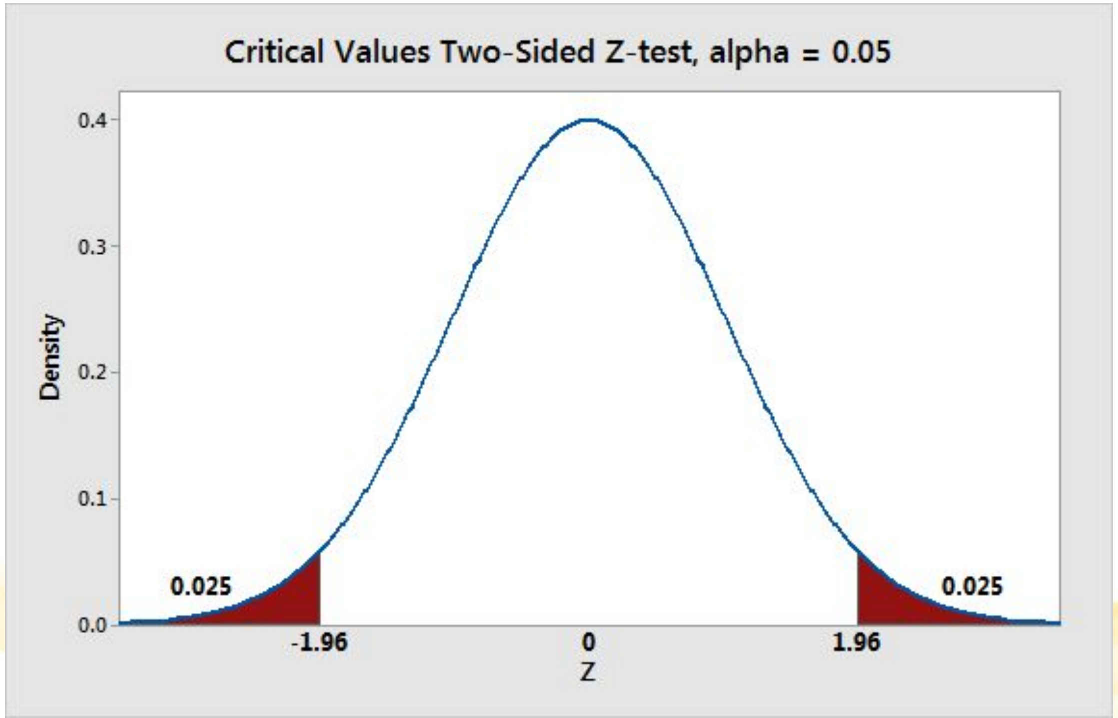
Two-tailed test

Two-sided hypothesis tests

In a 2-sided test, the null hypothesis is rejected if the **test statistic is too small or too large**.

- The rejection region for such a test consists of two parts: one on the left and one on the right.

Example: Test whether a sample mean is significantly different from the population mean, in either direction.

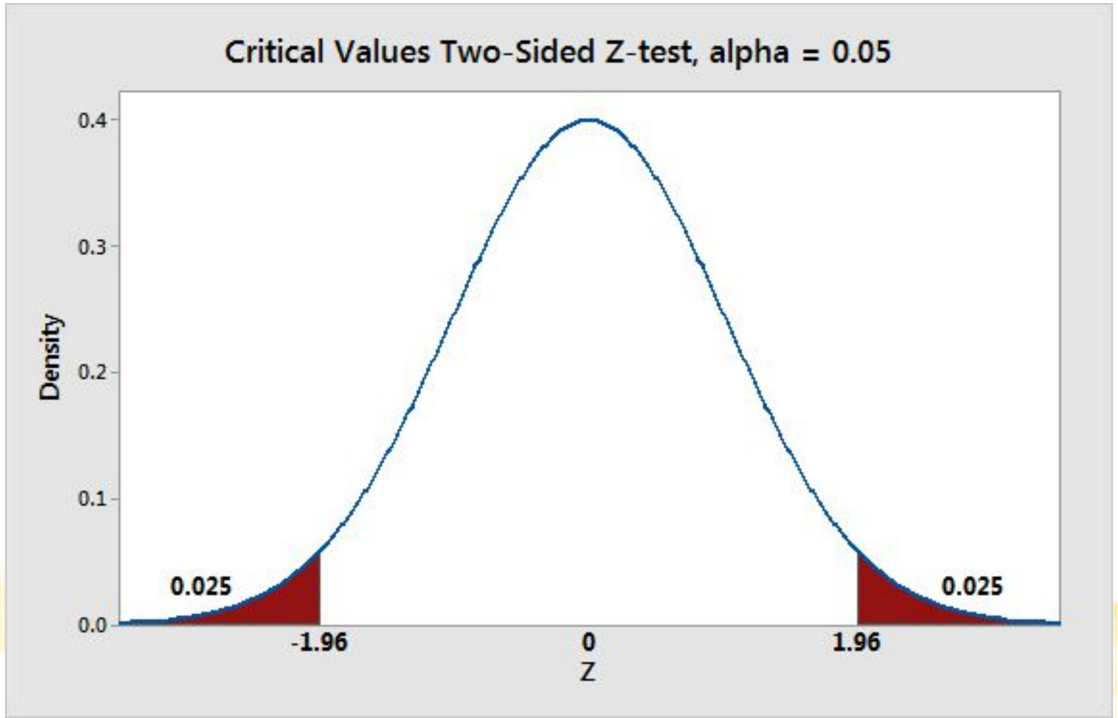


H_0: $\mu = \mu_0$
H_a: $\mu \neq \mu_0$

Two-sided hypothesis tests

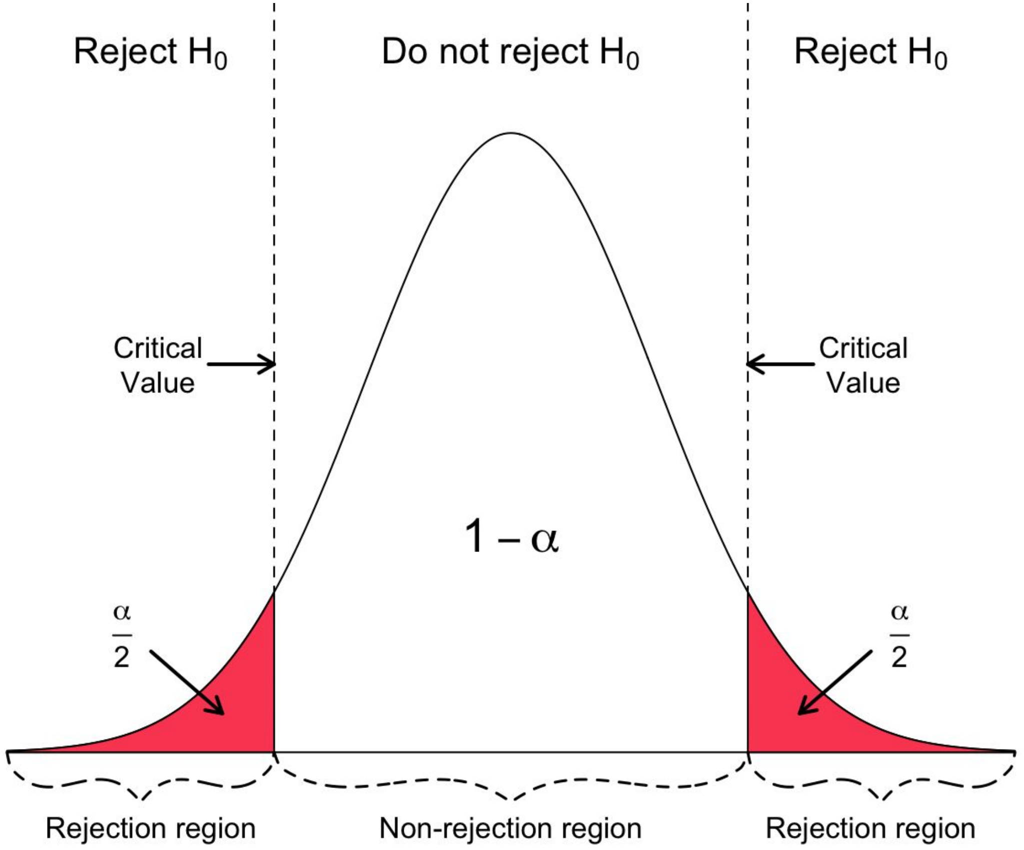
Because two-sided hypothesis tests have **two rejection regions**:

- You need **two critical values** to define them.
- You must **split the significance level in half**
  - Each rejection region has a probability of $\alpha/2$, making the total likelihood for both areas equal the significance level.



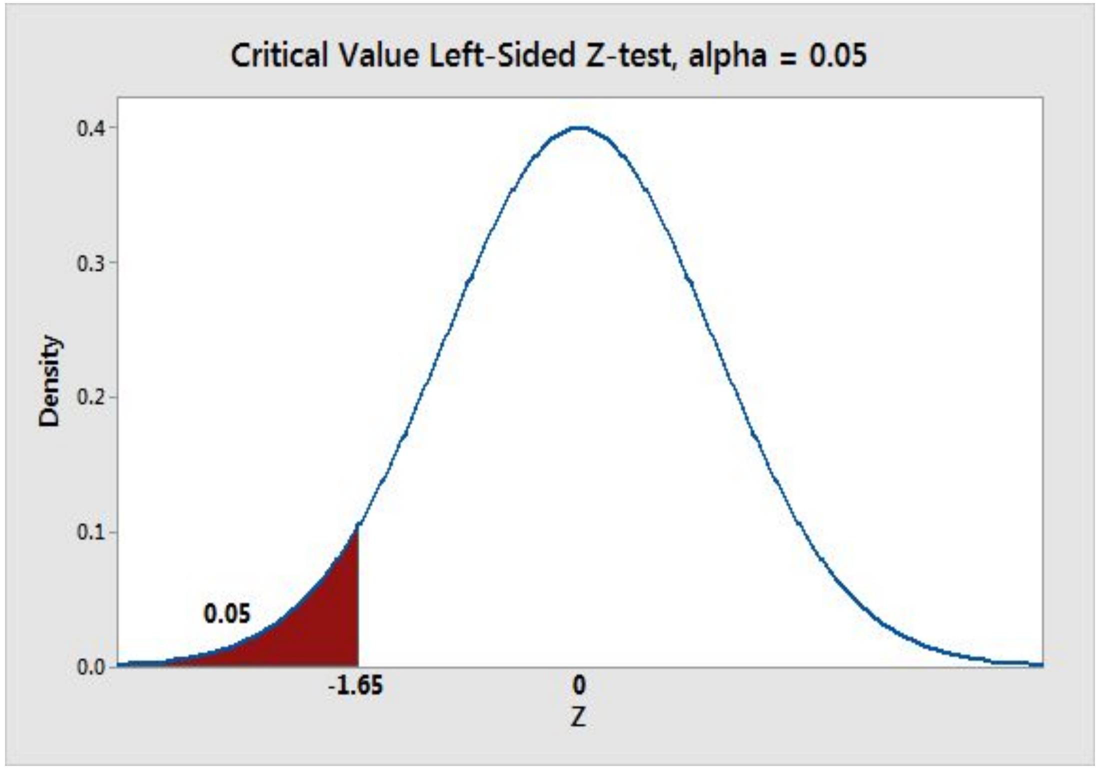
Two-sided hypothesis tests

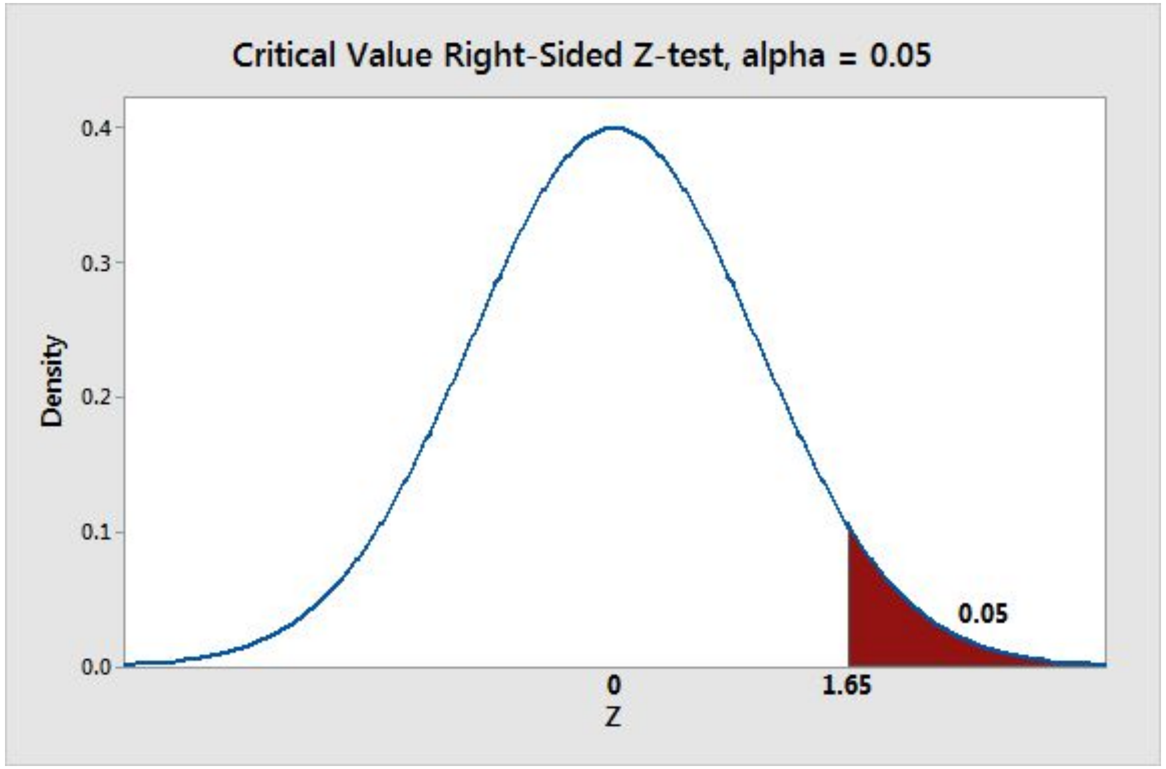
- The critical area of the distribution is two-sided, and tests **whether a sample is greater than or less than** a certain range of values.
- If the sample being tested falls into either of the critical areas, the alternative hypothesis is accepted and the **null hypothesis is rejected**.



One-Sided Tests

One-tailed tests have one rejection region, and thus only one critical value. The total α probability goes into that side.





One-Sided Tests

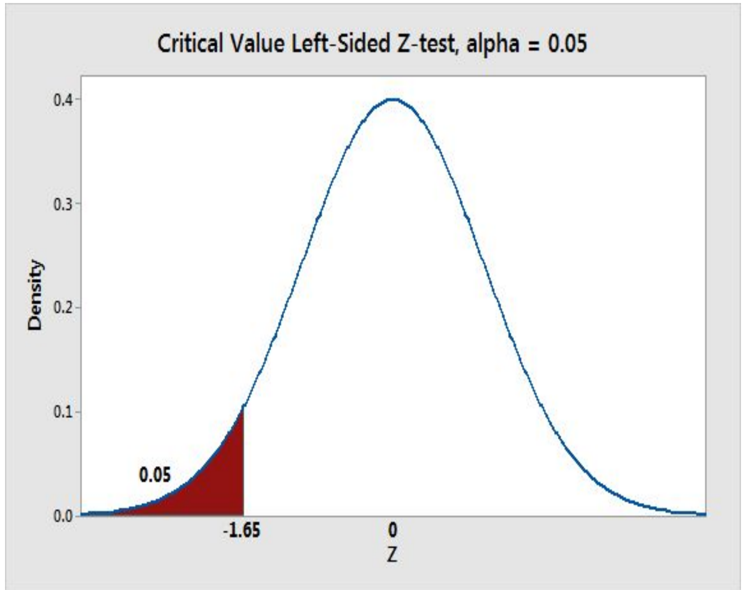
Left-Tailed Test: The rejection region is located to the extreme left of the distribution.

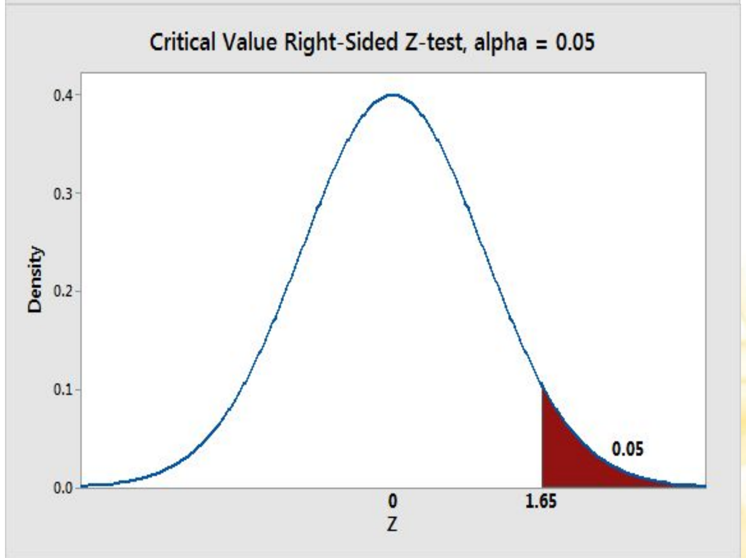
- Ex. H_a contains the condition: $\mu < \mu_0$

Right-Tailed Test: The rejection region is located to the extreme right of the distribution.

- Ex. H_a contains the condition $\mu > \mu_0$

H_0: $\mu \geq \mu_0$ H_a: $\mu < \mu_0$



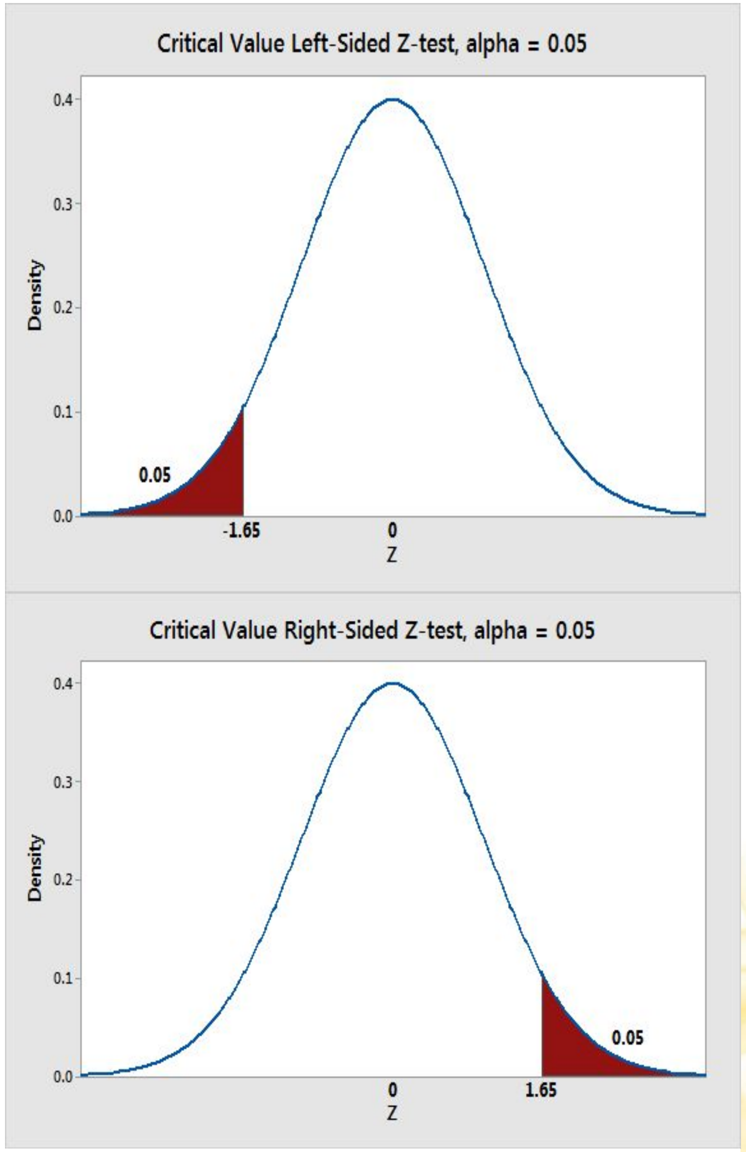


H_0: $\mu \leq \mu_0$ H_a: $\mu > \mu_0$

One-Sided Tests

- The critical area of the distribution is one-sided so that it is ***either* greater than or less than a certain value**, **but *not both***.
- If the sample being tested falls into the one-sided critical area, the alternative hypothesis is accepted and **the null hypothesis is rejected**.

$H_0: \mu \geq \mu_0$ $H_a: \mu < \mu_0$



$H_0: \mu \leq \mu_0$ $H_a: \mu > \mu_0$

How confident are you with your decision?

Understanding Level of Confidence and Significance

How confident are we in our decision?

Confidence Level (C): Represents the degree of certainty in our decision-making process. Ex. could be 95%, 99%.

Level of Significance (α): Indicates the likelihood of making a Type I error, which is rejecting the null hypothesis when it's true (a false positive).

- α is a term used in scientific research to describe when to reject the null hypothesis.
  - Typically $\alpha = 5\%$ or 1% (usually $0 \leq \alpha \leq 0.05$)
- Gives us a quick way of describing whether or not it is likely an effect exists.

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$$\alpha = 1 - C$$

Understanding Level of Confidence and Significance

How confident are we in our decision?

Interpreting Confidence Level:

- **High Confidence Level (ex. 99%)**
  - If we're 99% confident in our decision and reject the null hypothesis, we're highly certain that rejecting the null is the correct choice.
- **Low Confidence Level (ex. 50%)**
  - If our confidence level is only 50%, we're less convinced and might hesitate to reject the null hypothesis.

Understanding Level of Confidence and Significance

How confident are we in our decision?

Remember: **$\alpha = 1 - C$** (always sum up to 1)

Ex. "Level of Confidence"=95%

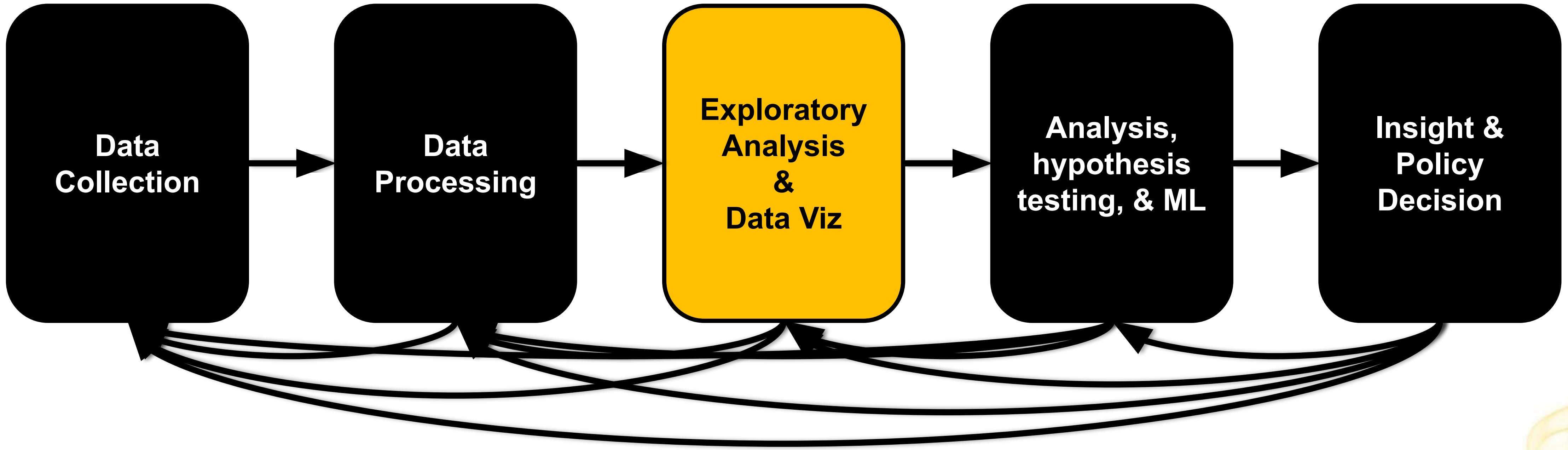
- $C = 0.95$
- $\alpha = 1 - C$
  $= 1 - 0.95 = 0.05 = 5\%$

Note: C and α are often used **interchangeably.**

Both convey the same concept:

How certain we are about the correctness of our decision.

Data Visualization & Storytelling


Data Collection
Data Processing
Exploratory Analysis & Data Viz
Analysis, hypothesis testing, & ML
Insight & Policy Decision

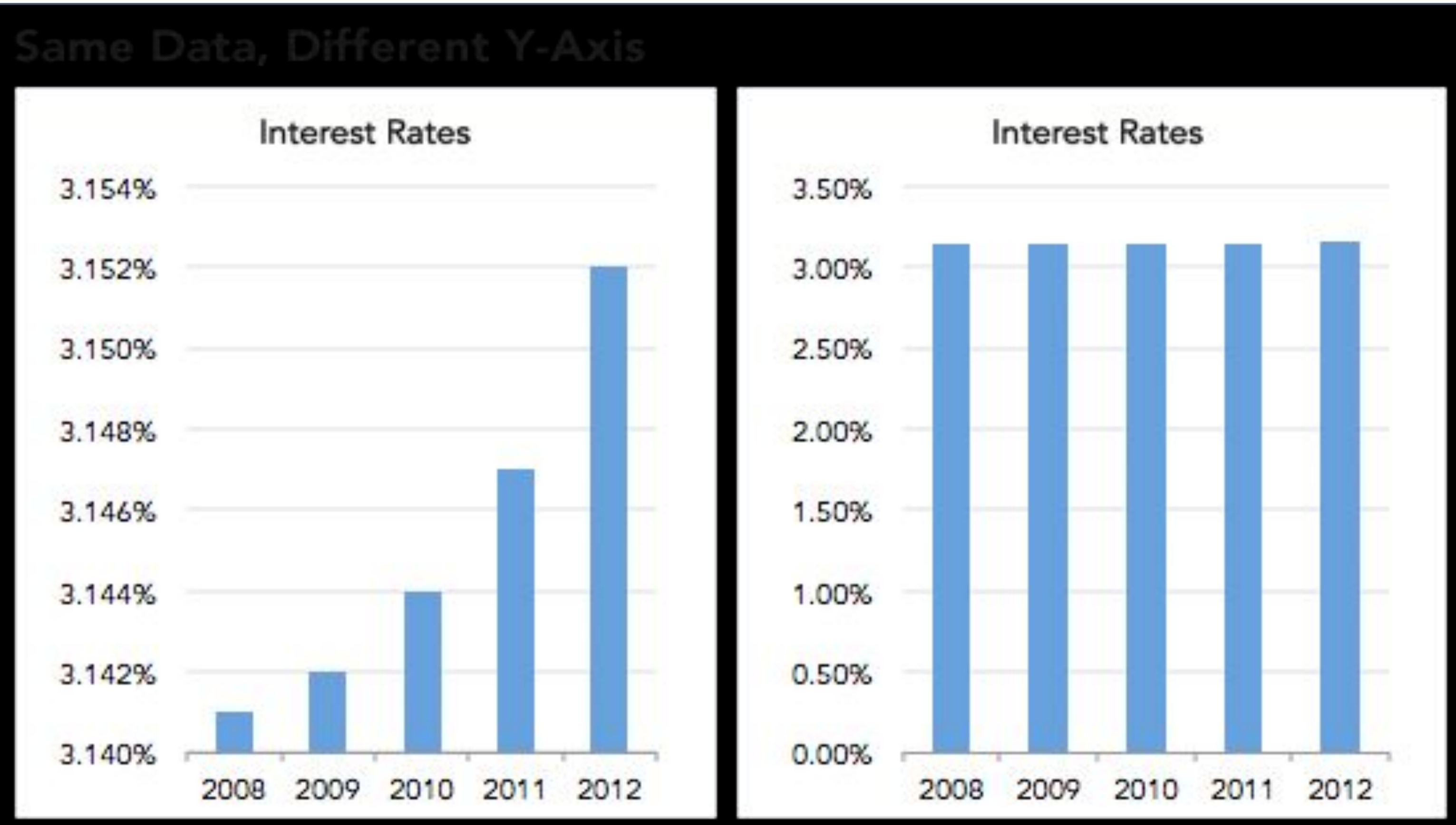
What is Data Visualization?

The process of **creating graphical representations of information**.

- **Represent** data and information graphically.
  - Translate data into a visual context using charts, plots, heatmaps, animations, infographics, etc.
- **Discover** the trends and patterns of data.
- **Summarize** the information

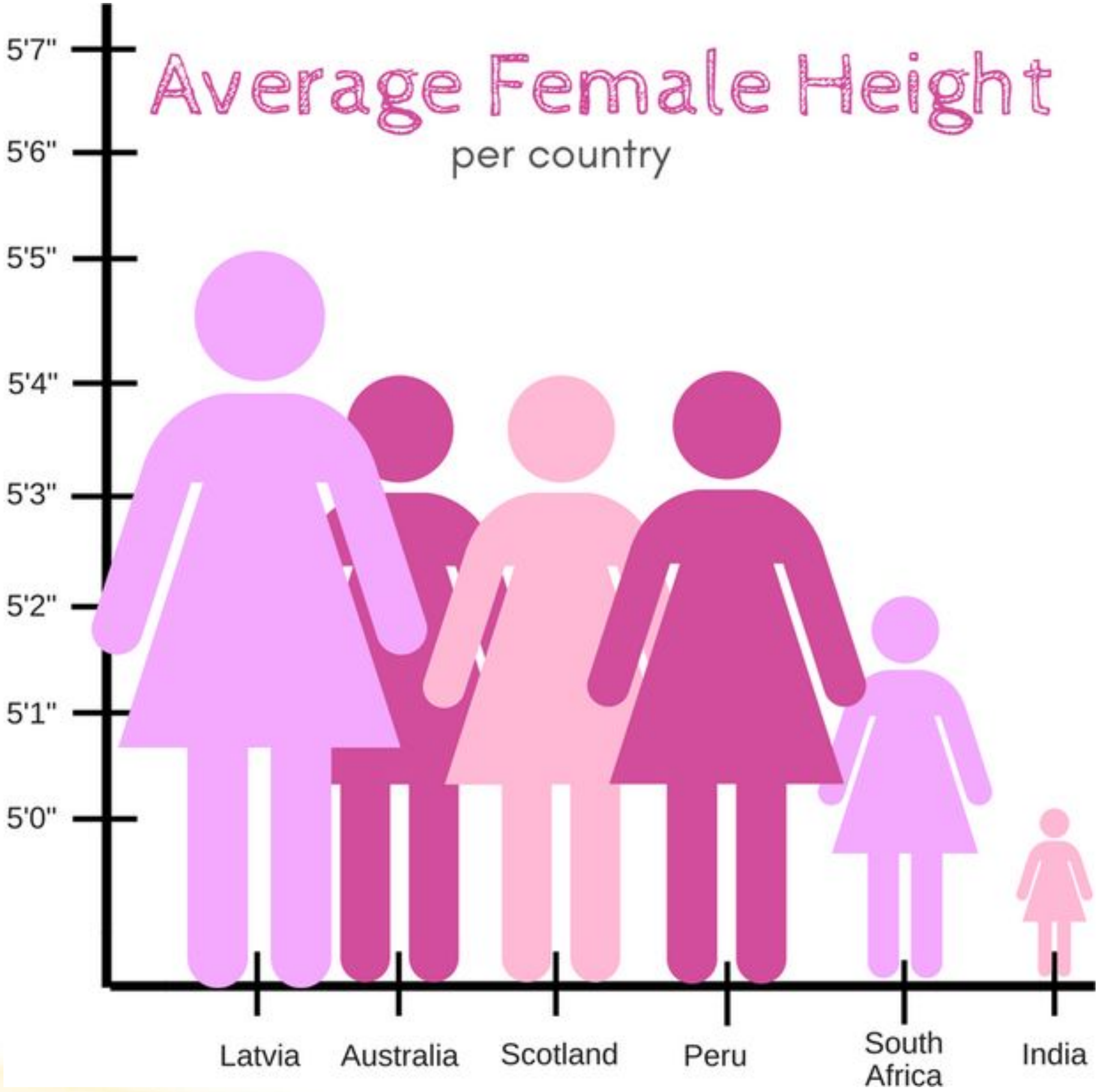
The Bad

Sometimes, visualizations can be misleading



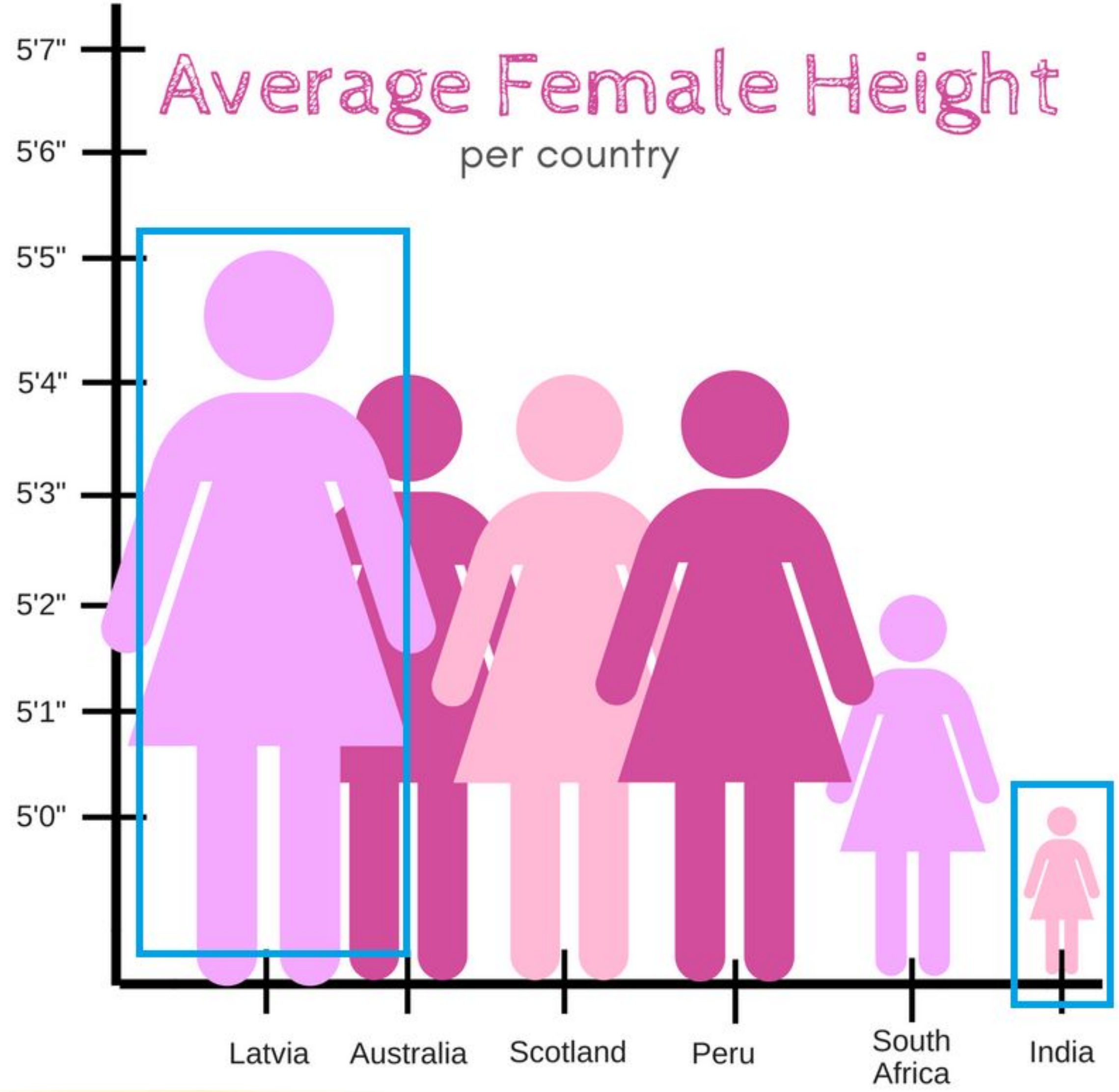
Is this different data?

Images Source Tumblr



What's wrong with this graph?

Images Source Tumblr

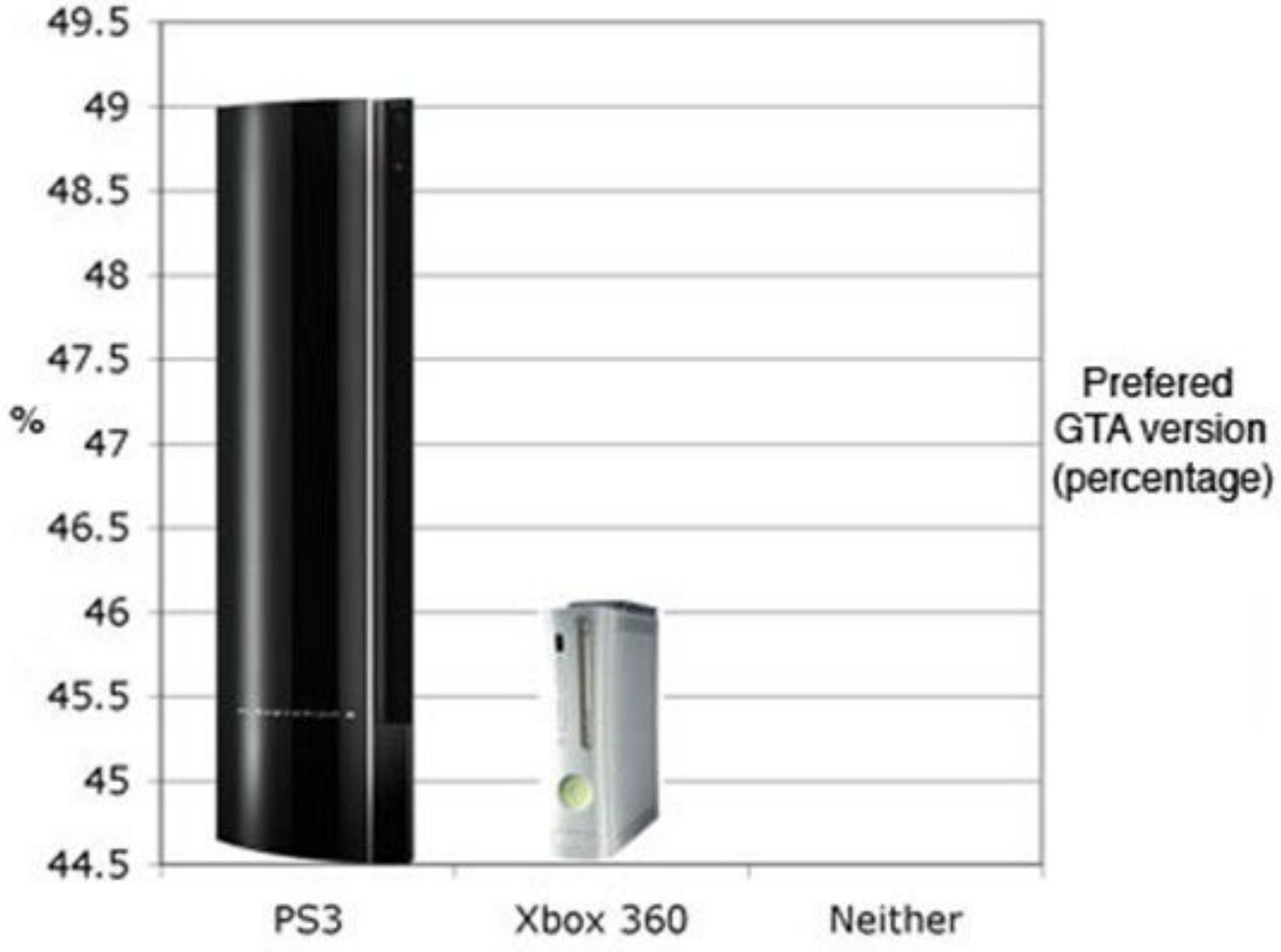


What's wrong with this graph?

Actual difference – 0.5 feet. Visual Difference – 10 feet.

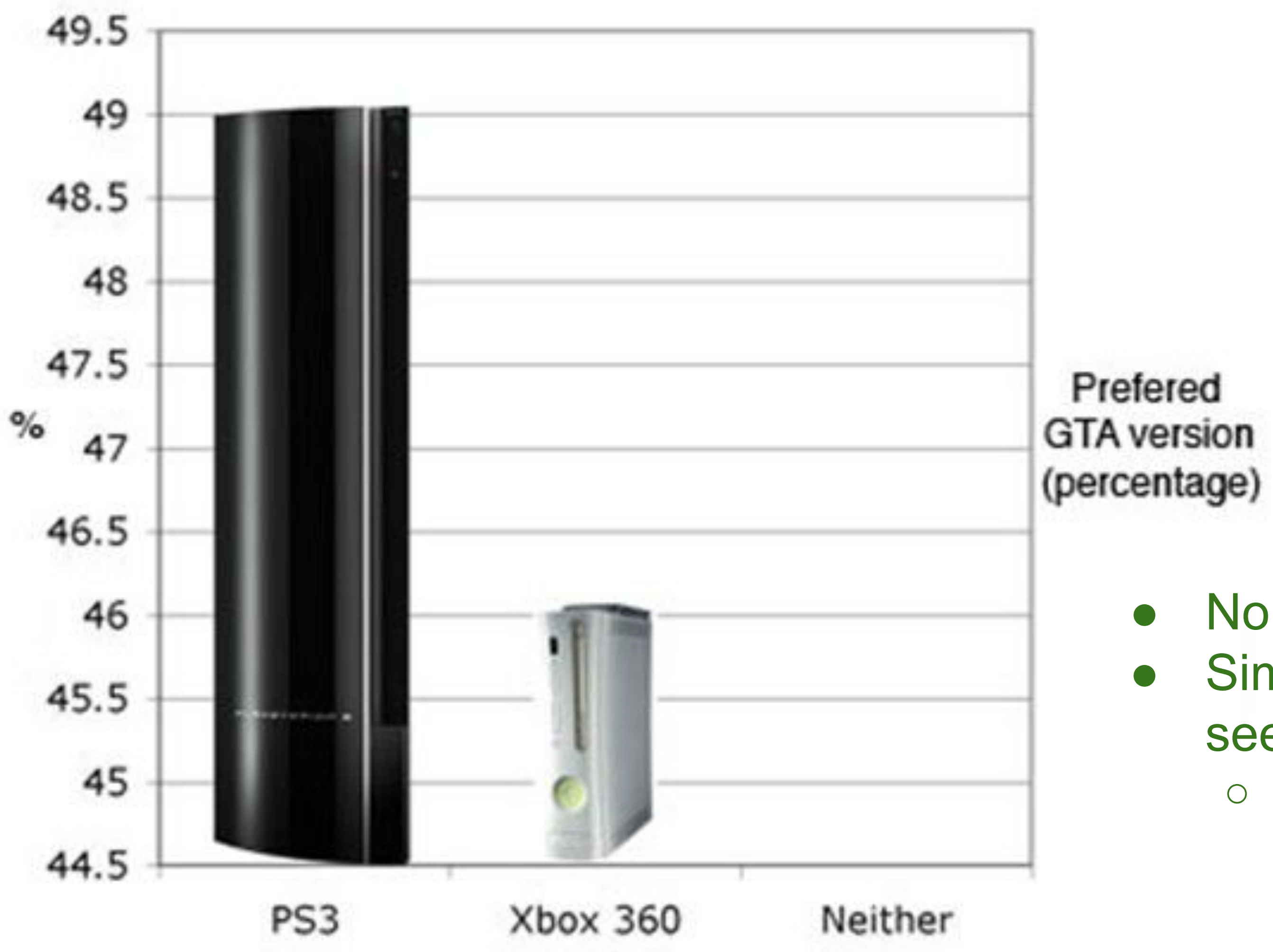
Visualizations should never prioritize style over substance.

A good visualization needs to be arranged carefully and in a logical manner



What's wrong with this graph?

Source: https://slideplayer.com/slide/9738183



What's wrong with this graph?

- Non-zero baseline inflates the y axis
- Similar to height graph, the image makes it seem like the sales are 3x
    - Reality is a 3% difference in sales

Source: https://slideplayer.com/slide/9738183

Common things poor visualizations do

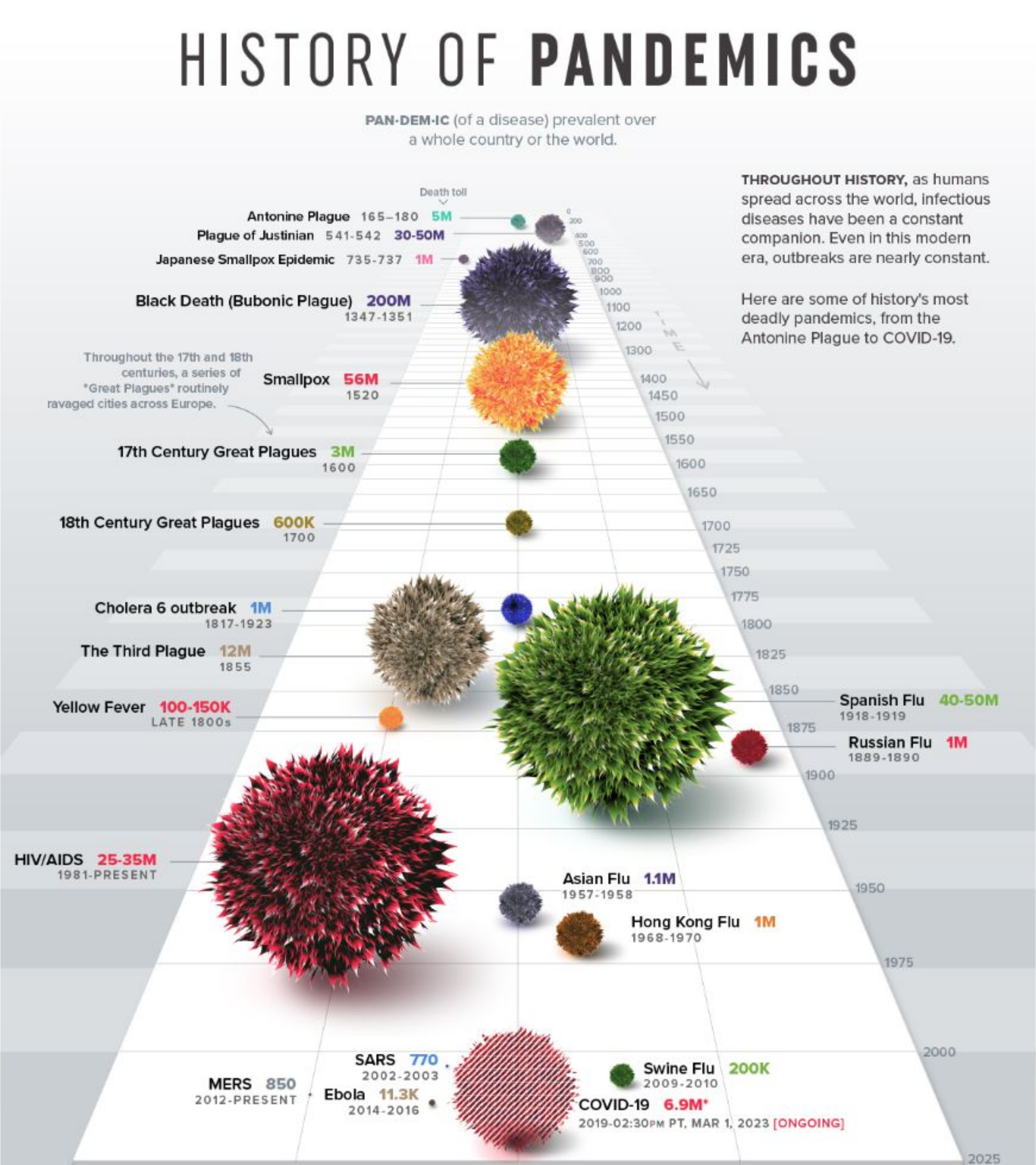
1. **Lack clarity**
   - ex. Putting too many variables together without making crisp visual delineations of why or how they can be understood together
   - ex. Making color choices that don’t aid in or actively harm understanding the visual
   - ex. Choosing red to represent “go” and green to represent “stop”, confusing the viewer
2. **Inappropriately change scale**
   - ex. Changing the scale of y axis to make differences seem larger or smaller
   - ex. Changing the scale a variable in a multivariable chart without making it clear that they have different scales
3. **Choose style over substance**
   - ex. Using images/pictograms that look nice but interfere with the correct understanding of the chart
   - ex. Using a 3D pie chart because it looks cool even though 2D is clearer
4. **Cherry Pick**
   - ex. Choosing only a subset of the data that supports a hypothesis or story you want to convey
5. **Outright Lie**
   - ex. Omitting key information that is pertinent to understanding the reality/truth of the data
   - ex. Making changes to the graph so that viewers believe something different from what the actual data shows

Takeaway

Be careful when designing visualizations, and

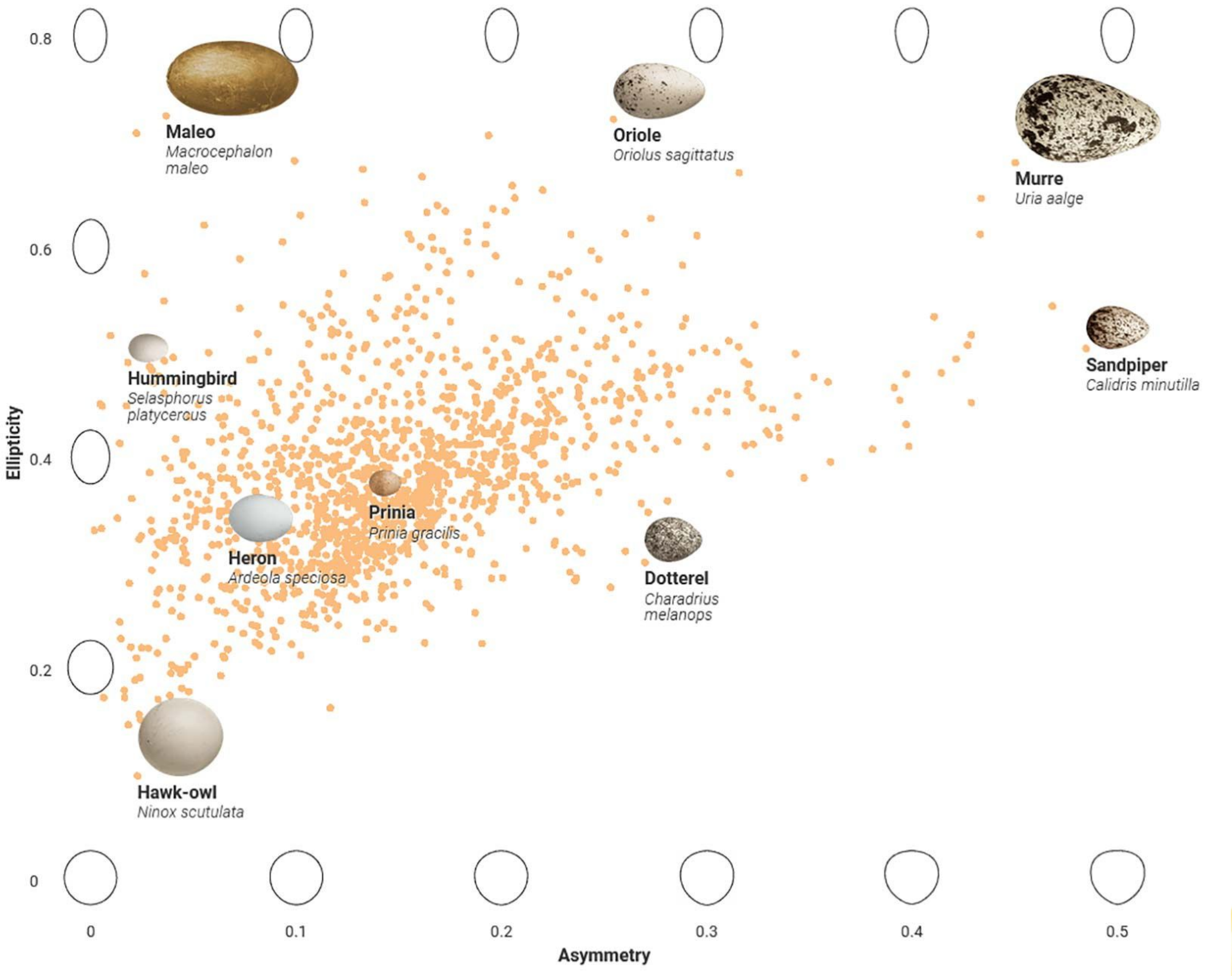
be extra careful when interpreting graphs created by others.

What's good or bad about this visualization?



Source: https://www.visualcapitalist.com/history-of-pandemics-deadliest

What’s good or bad about this visualization?



Source: https://vis.sciencemag.org/eggs

Interactive Examples

- Matt Korostoff - Pixel Wealth
  - https://mkorostoff.github.io/1-pixel-wealth
- NASA - Eyes on Asteroids
  - https://eyes.nasa.gov/apps/asteroids/#/home
- Shane Mielke - Launch It
  - https://launchit.shanemielke.com
- Washington Post - Solar Eclipses in Your Lifetime
  - https://www.washingtonpost.com/graphics/national/eclipse
- NY Times - How every member got to congress (US)
  - https://www.nytimes.com/interactive/2019/01/26/opinion/sunday/paths-to-congress.html

Why we visualize

Importance of Data Visualizations

Good data visualizations allow us to reason and think effectively about our data.

Presenting information visually allows us to offload internal cognition to the perceptual system, making it easier to understand and process.

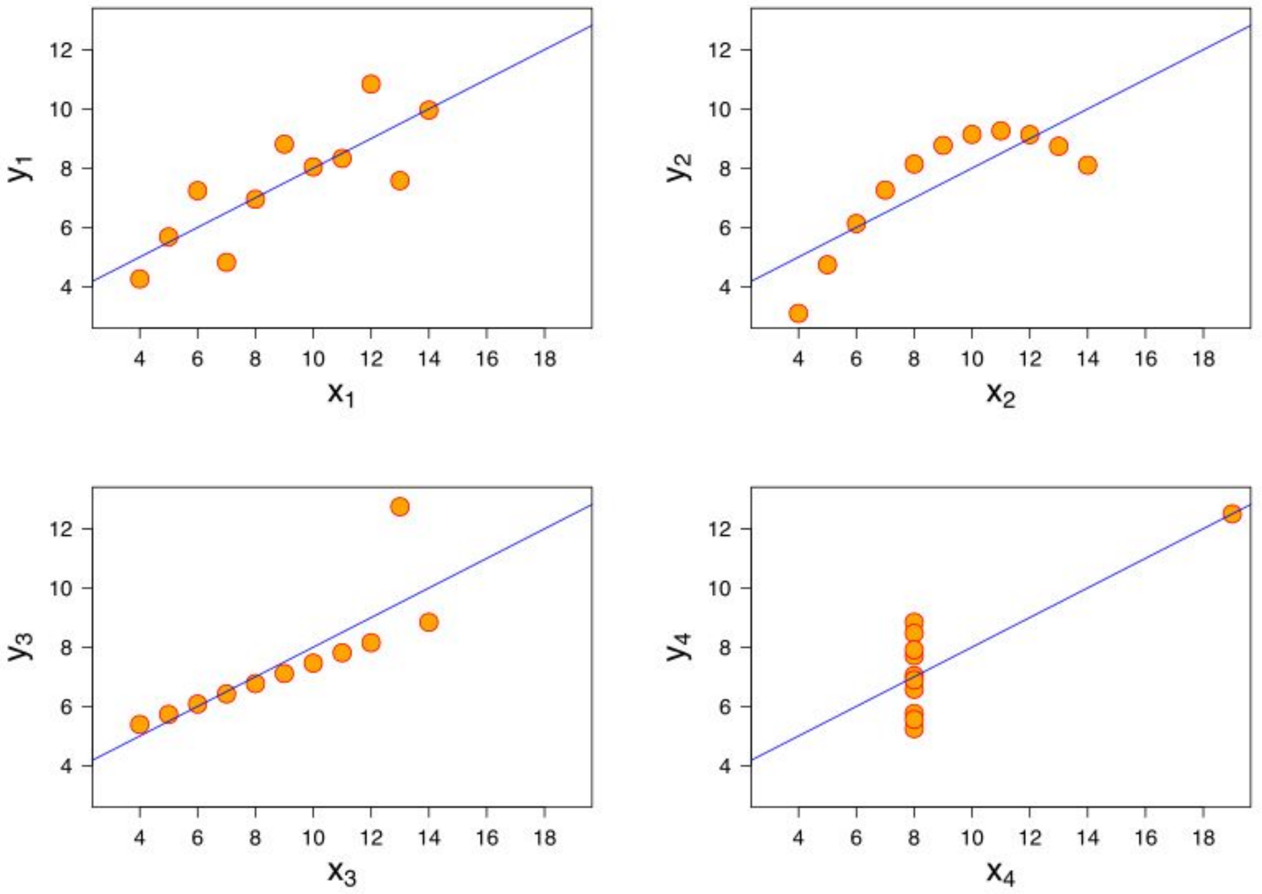
I		II		III		IV	
10	8.04	10	9.14	10	7.46	8	6.58
8	6.95	8	8.14	8	6.77	8	5.76
13	7.58	13	8.74	13	12.74	8	7.71
9	8.81	9	8.77	9	7.11	8	8.84
11	8.33	11	9.26	11	7.81	8	8.47
14	9.96	14	8.1	14	8.84	8	7.04
6	7.24	6	6.13	6	6.08	8	5.25
4	4.26	4	3.1	4	5.39	19	12.5
12	10.84	12	9.13	12	8.15	8	5.56
7	4.82	7	7.26	7	6.42	8	7.91
5	5.68	5	4.74	5	5.73	8	6.89

"**Anscombe's quartet**" (Francis Anscombe,1973):
4 datasets that share the same descriptive statistics, including mean, variance, and correlation (by variable).

Importance of Data Visualizations

When we graphically represent them, the patterns are clearly different.

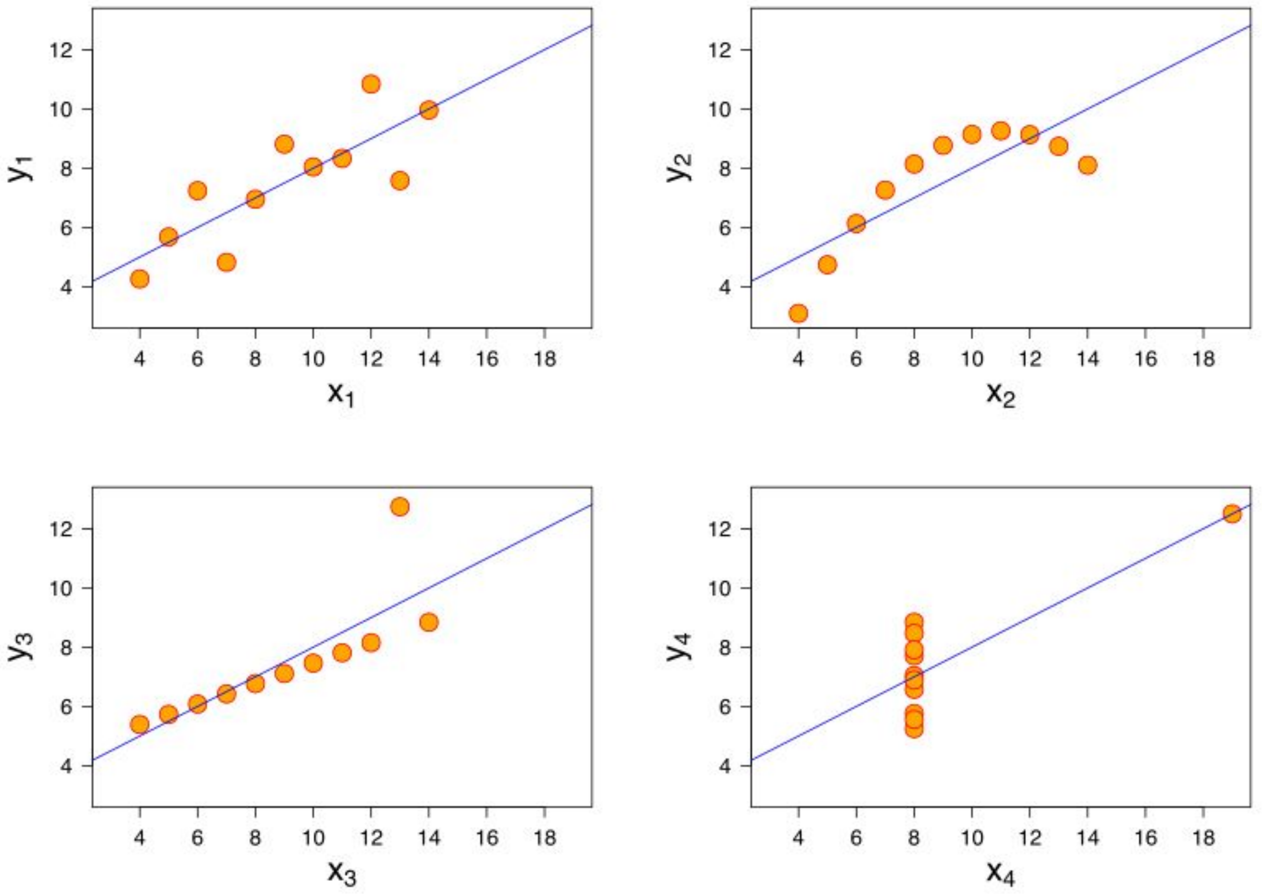
<u>This is the power of effective data visualization</u>: it allows us to bypass cognition by communicating directly with our perceptual system (understand things quickly without having to think too hard).



Why we create visualizations

Think about how **effortlessly** your visual cortex understood the **complex** mathematical relationships in these data sets.

The occipital lobe is highly optimized, especially compared to newer inventions like language and the frontal lobe (which does traditional “math”).



Who do we create visualizations for?

- For **ourselves**:
  - When we want to learn about and explore our data
  - When we want to investigate a specific hunch

Who do we create visualizations for?

- For **ourselves**:
  - When we want to learn about and explore our data
  - When we want to investigate a specific hunch
- For **others**:
  - When we want to communicate something to other people
    - To inform
    - To persuade
    - To educate
    - To engage

Why Data Visualization

Data can be difficult to interpret when presented in a solely textual/numerica manner (ex. CSVs, tables)

Using graphical/visual representations of information, data visualization presents complex data in **easy-to-understand** ways, allowing users to **quickly discover** essential patterns, trends, and anomalies.

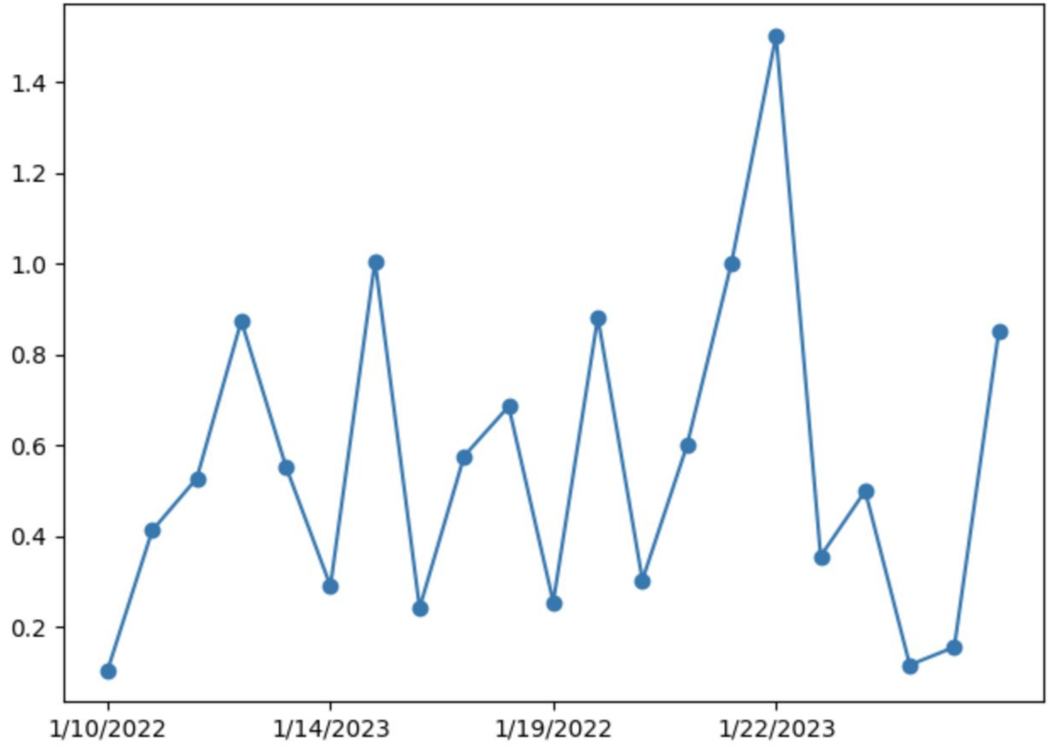
Types of Visualizations

(and when to use them)

Line charts: display trends

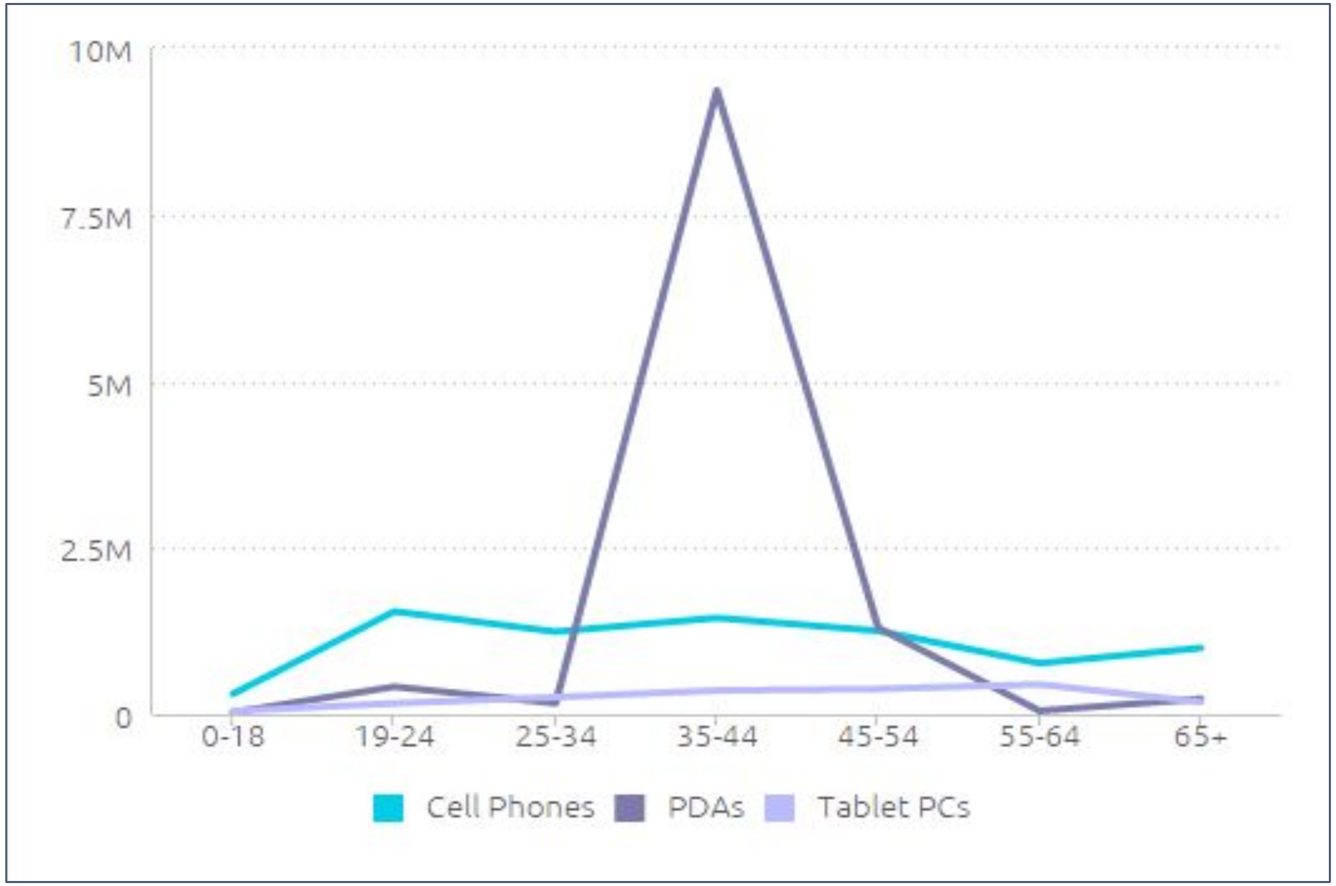
A line graph reveals **trends or progress over time.**

- Show trends across categories for easy comparison.
- Used when we want to chart a continuous data set.




```
plt.plot(sales['Date'], sales['Total Sold'], marker='o')
plt.xticks(np.arange(0, 20, 5))
plt.tight_layout()
plt.show()
```

Line Chart Example



Observe:

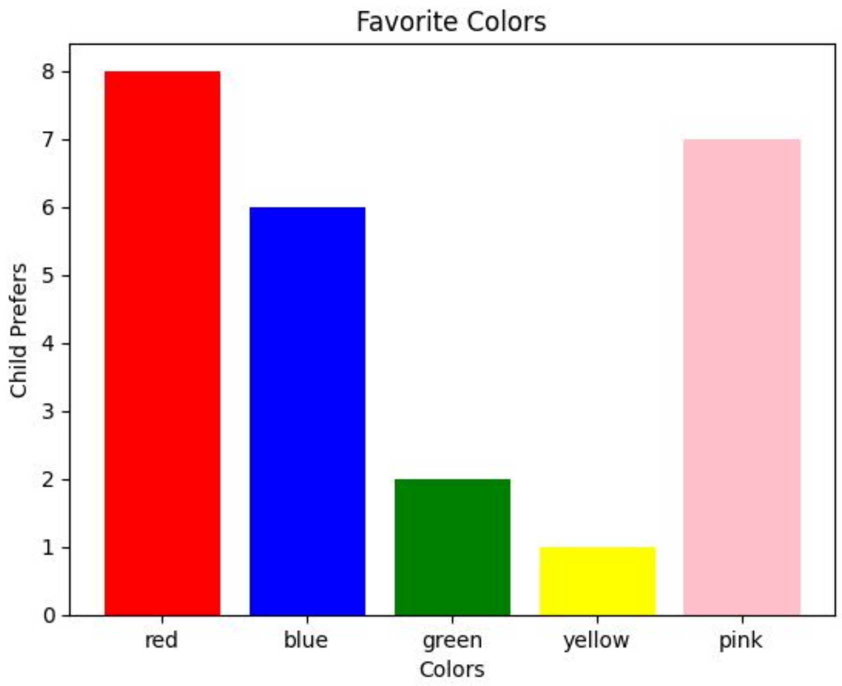
- Biggest customers are 35-44 year old buyers of PDAs
- Followed by 19-24 year old buyers of cell phones.

Ex: Sales figures by age group for three different product lines

Bar Charts: Break things down, simply

Bar graphs can help **compare data between different groups** or track changes over time.

- Most useful when there are **big changes** or when **comparing one group against another**.

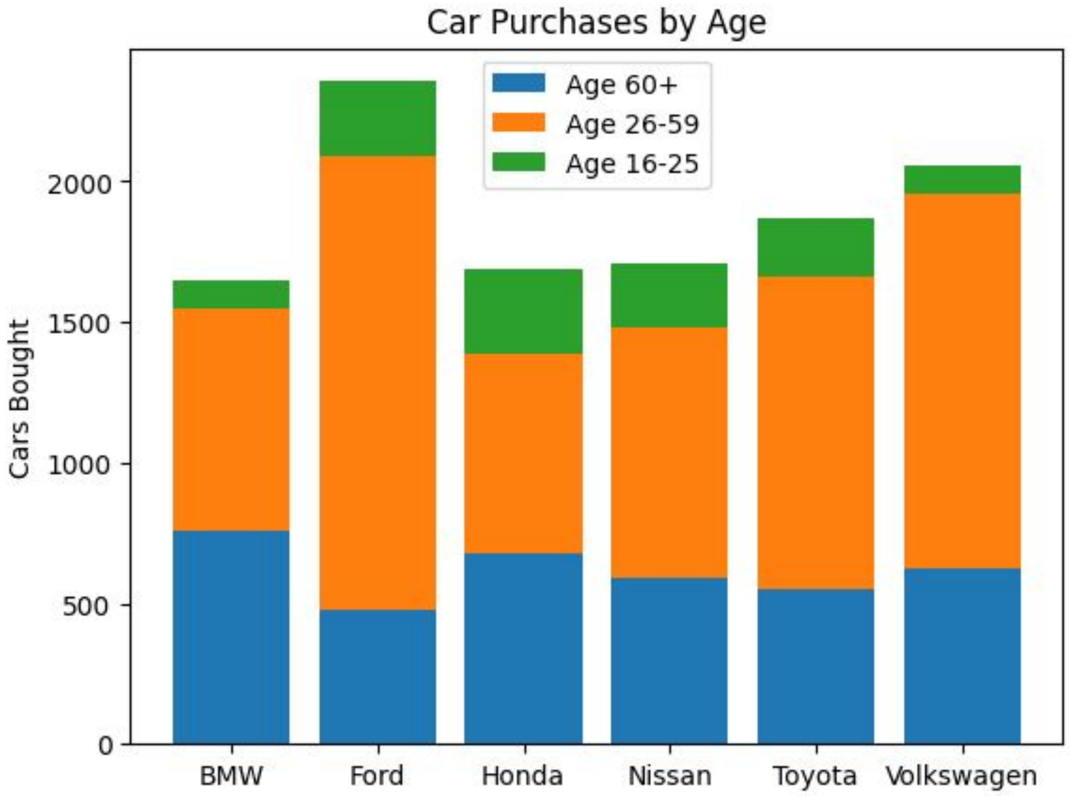



```python
fig, ax = plt.subplots()
ax.bar(x, y, color=colors)
plt.xlabel('Colors')
plt.ylabel('Child Prefers')
plt.title('Favorite Colors')
plt.show()
```

Stacked Bar Charts

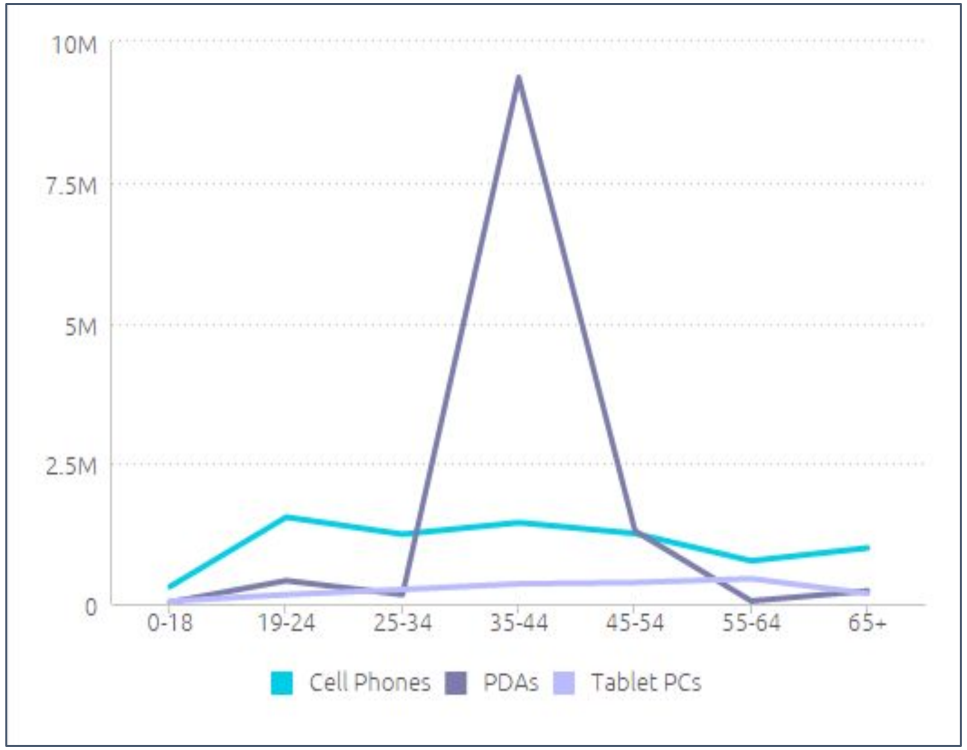
Stacked bar graphs allow you to show how a particular bar breaks down by category.

- Great for comparing several different categories, especially when breaking into color-coded categories helps to show patterns

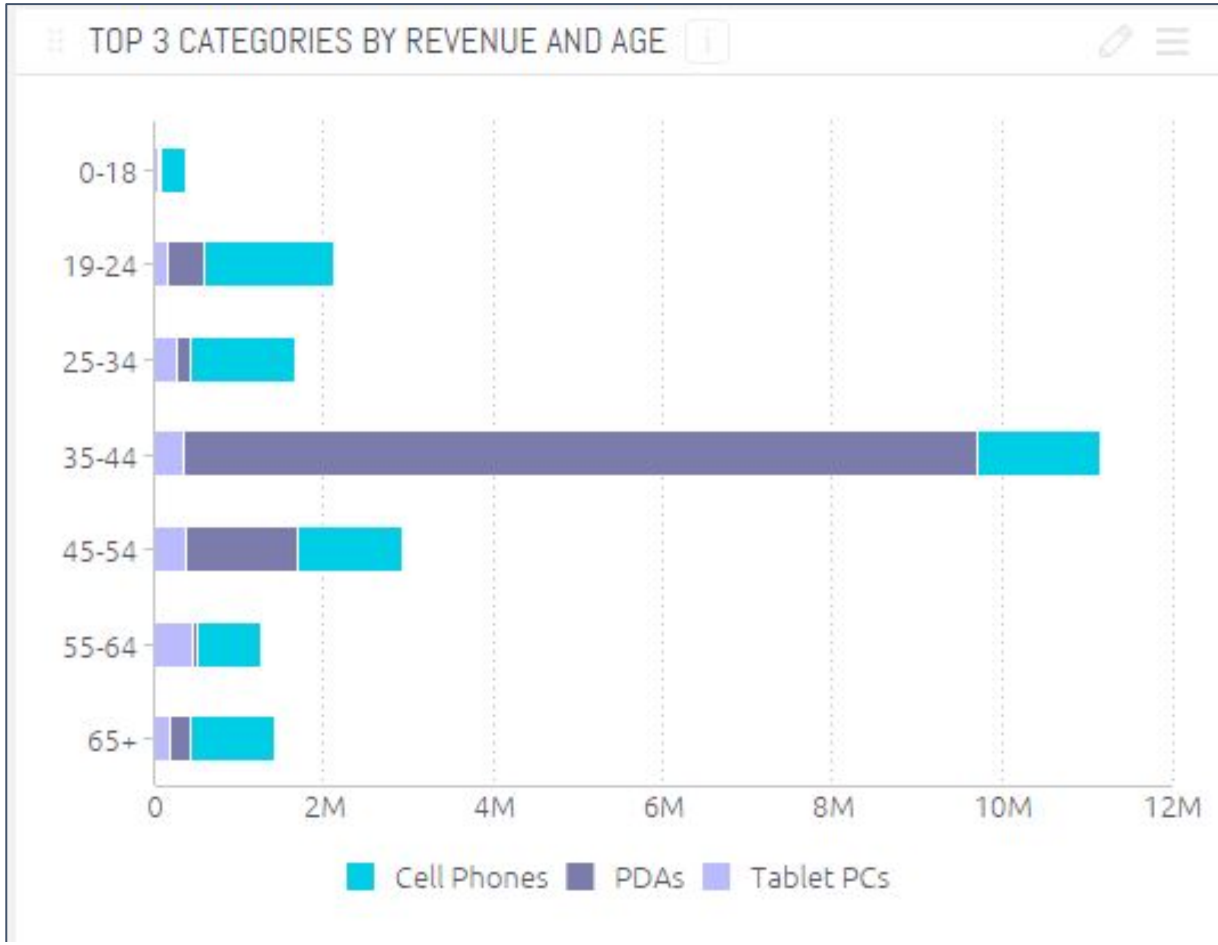



```
fig, ax = plt.subplots()
ax.bar(car_makers, oldest_buyers, label='Age 60+')
ax.bar(car_makers, middle_buyers, bottom=oldest_buyers, label='Age 26-59')
ax.bar(car_makers, young_buyers,
       bottom=np.array(oldest_buyers) + np.array(middle_buyers),
       label='Age 16-25')

ax.set_ylabel('Cars Bought')
ax.set_title('Car Purchases by Age')
ax.legend()
plt.show()
```

Bar Charts Example



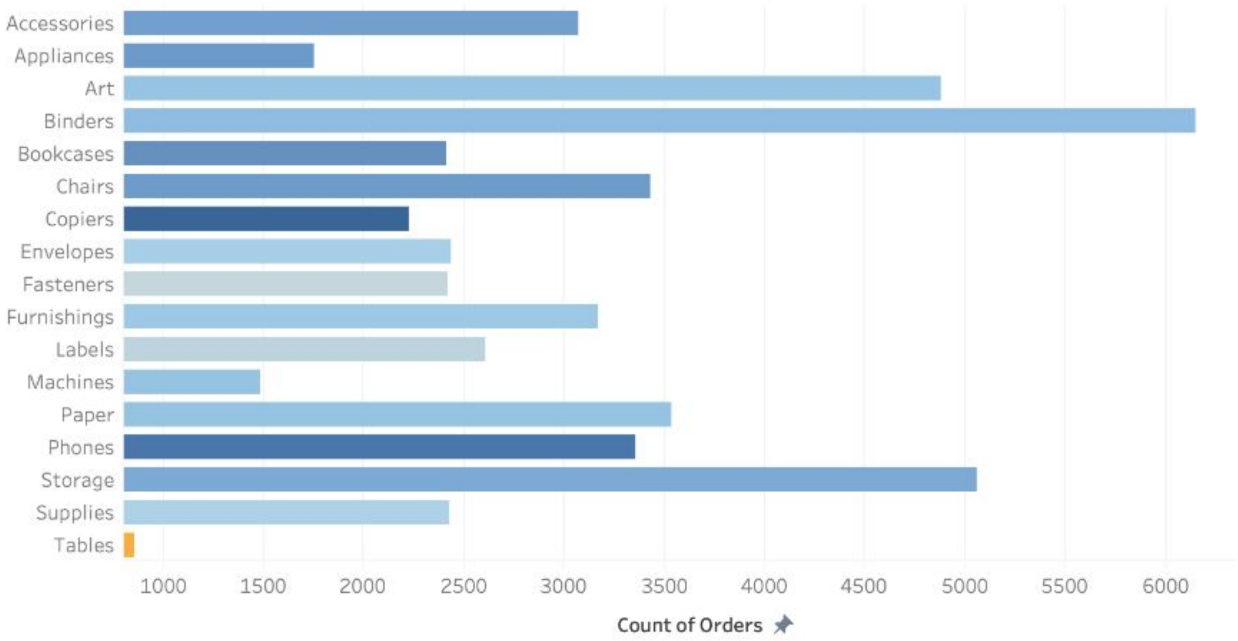
Revisualize previous chart as a stacked bar chart

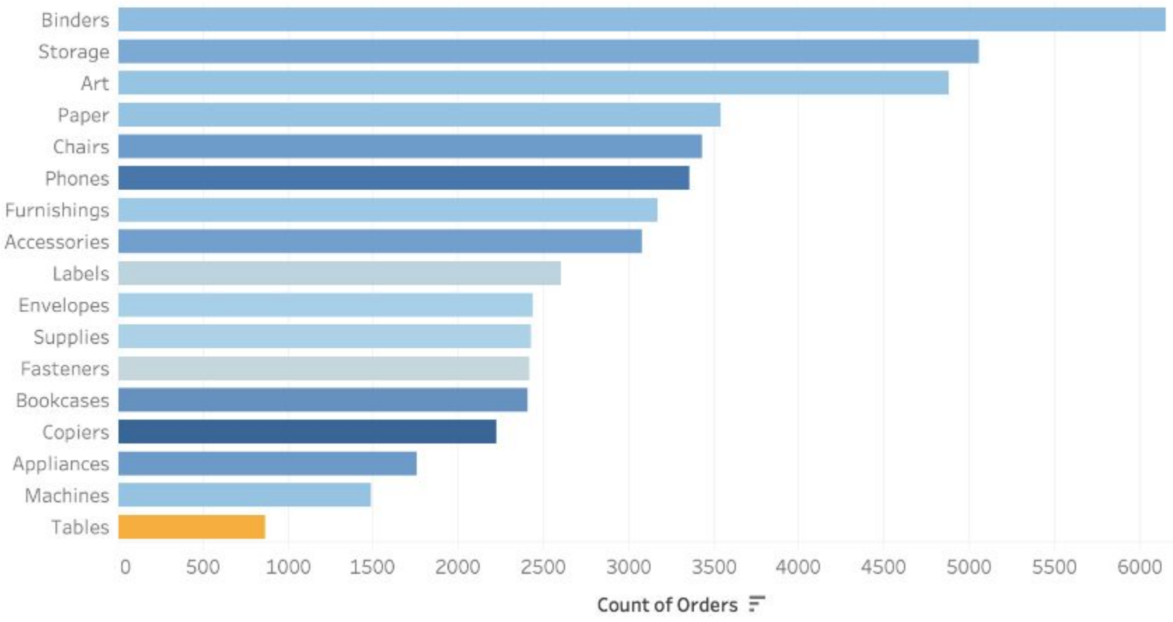


- Products are grouped by age
- Explore detailed sales differences within each age category.
- Quickly identify the most valuable age groups for business

Bar Charts Example

Which of these is better?



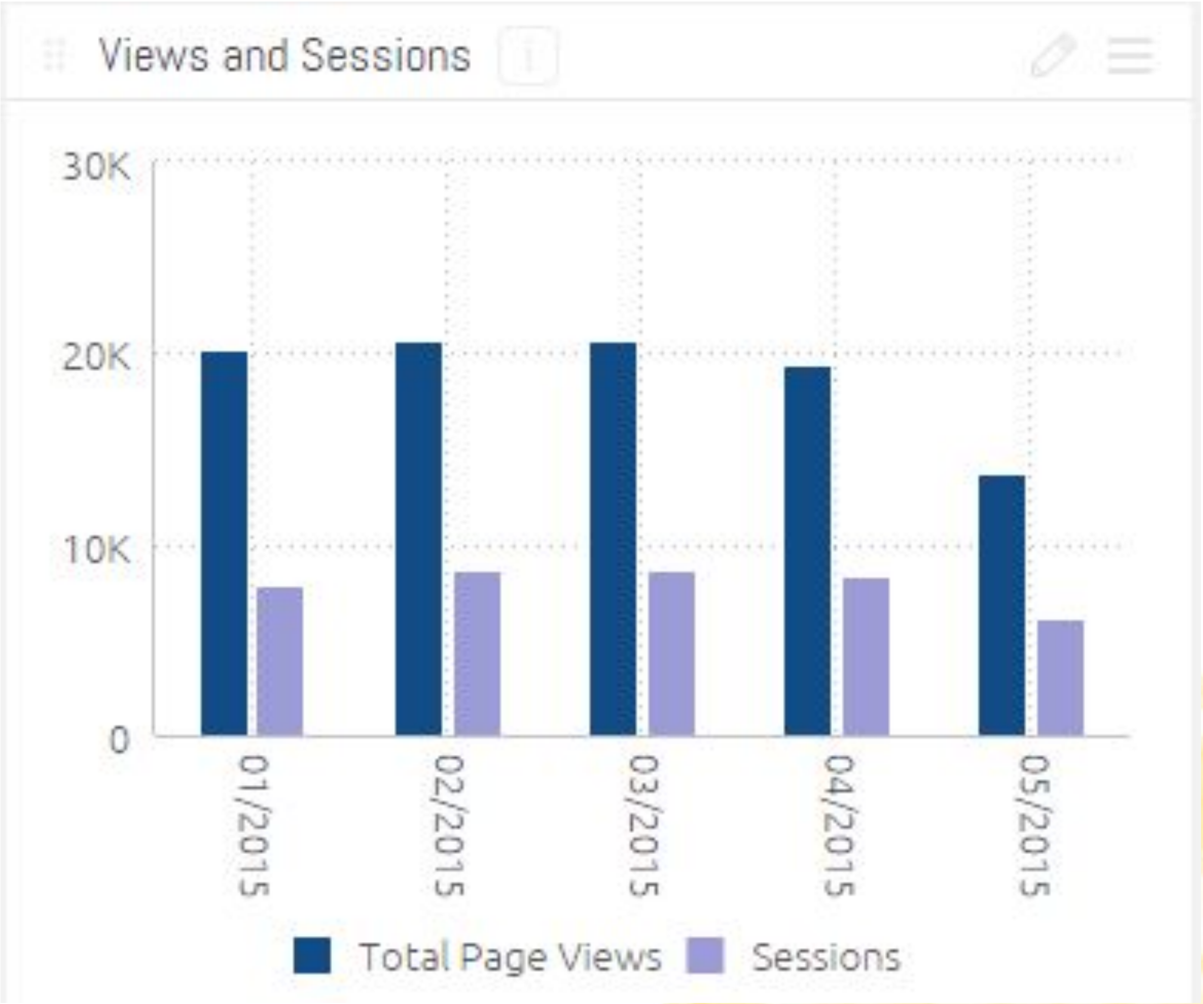


Source: https://www.tableau.com/data-insights/reference-library/visual-analytics/charts/bar-charts

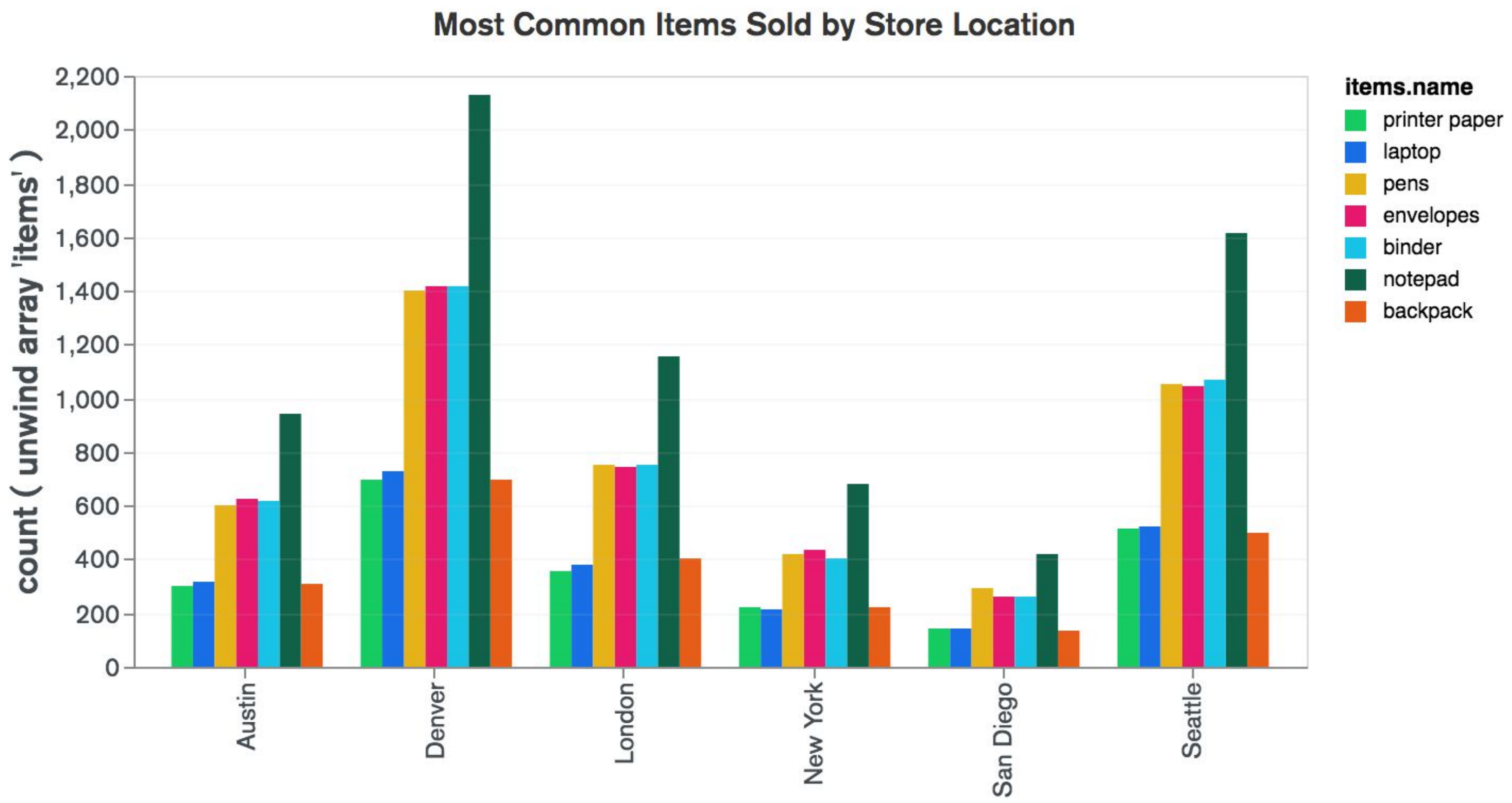
Column Charts: Compare values side-by-side

Useful for **side-by-side comparisons of different categories.**

- Show changes over time
- Good for scenarios where you want to compare and don't need to show parts of a whole



Column Charts Example

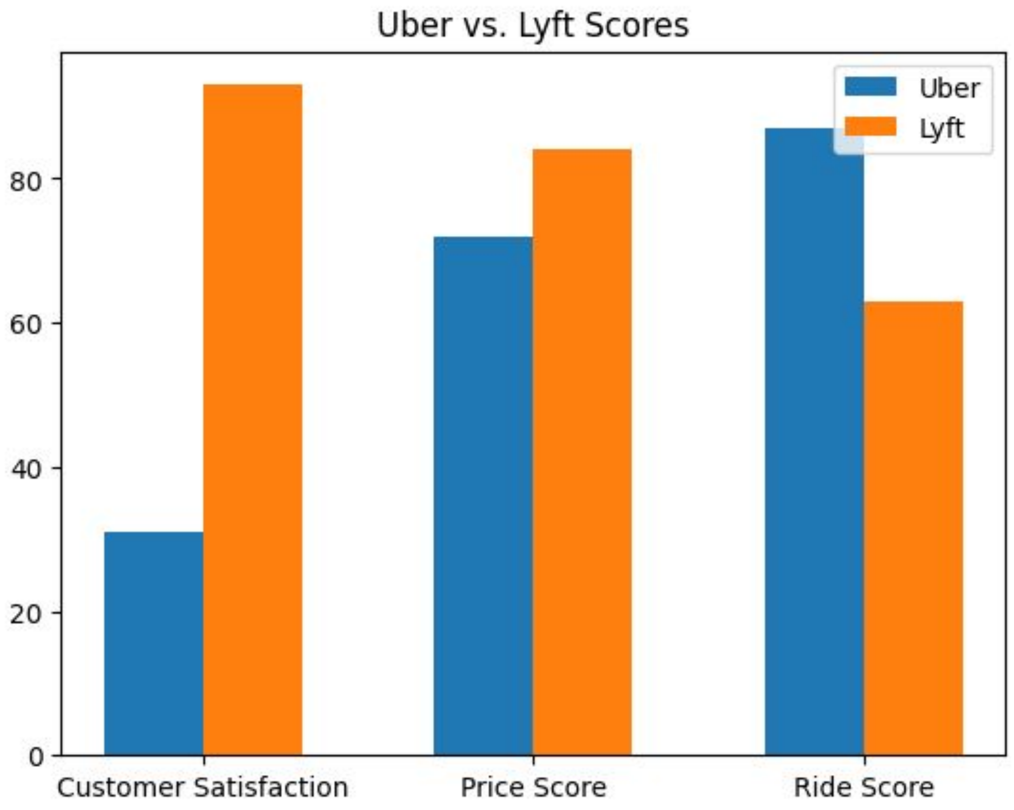


Source: https://forum.knime.com/t/grouped-bar-chart/33602

Use for side-by-side comparisons of different values.

- Easily see how popular each item is by location

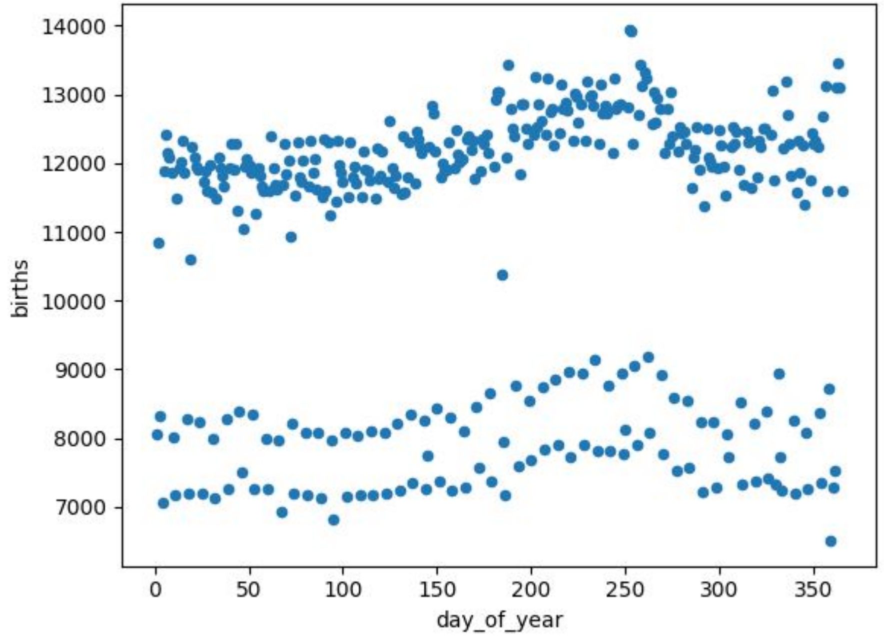
Column Charts Example




```python
x_labels = ['Customer Satisfaction', 'Price Score', 'Ride Score']
x_pos = np.arange(len(x_labels))
bar_width = 0.3

fig, ax = plt.subplots()

ax.bar(x_pos - bar_width/2, uber_vals, bar_width, label='Uber')
ax.bar(x_pos + bar_width/2, lyft_vals, bar_width, label='Lyft')

ax.set_xticks(x_pos)
ax.set_xticklabels(x_labels)
plt.legend()
plt.title('Uber vs. Lyft Scores')
plt.show()
```

Scatter Plots: Relationships

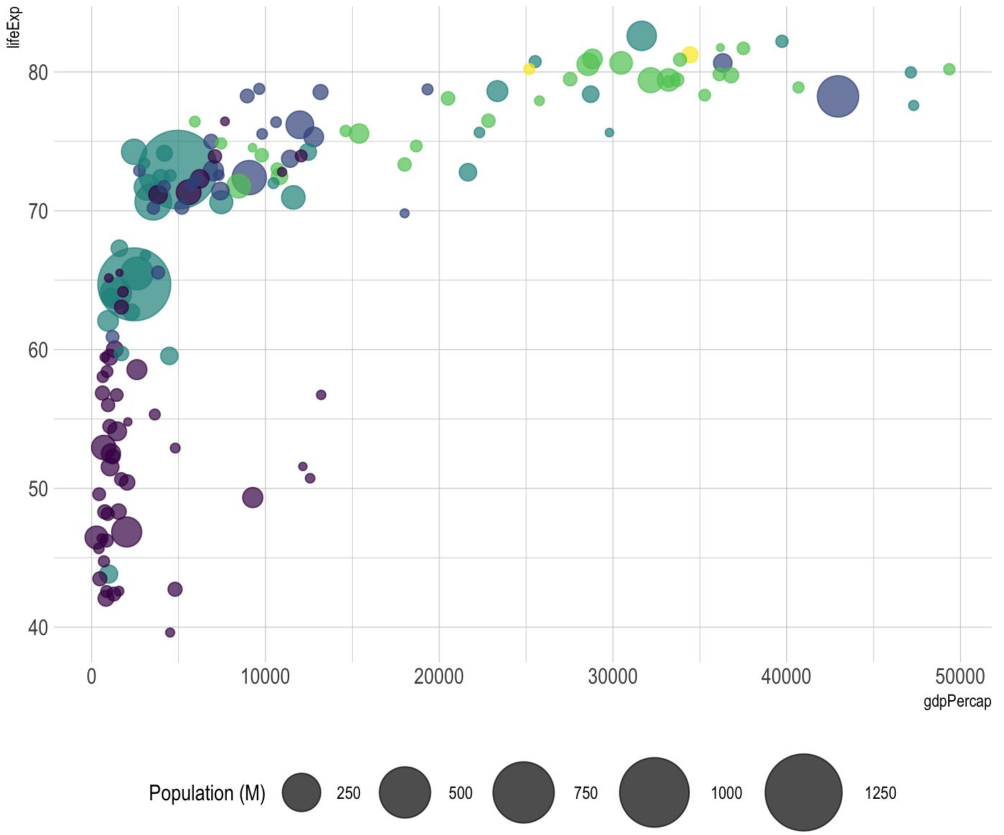
- Displays the relationship between variables
  - Usually 2 continuous variables
- Involves plotting individual data points as dots on a graph
  - One variable on the x-axis
  - The other variable on the y-axis.
  - Each dot represents an observation in the dataset
- The pattern formed by the dots can reveal the nature and strength of the relationship between the variables.



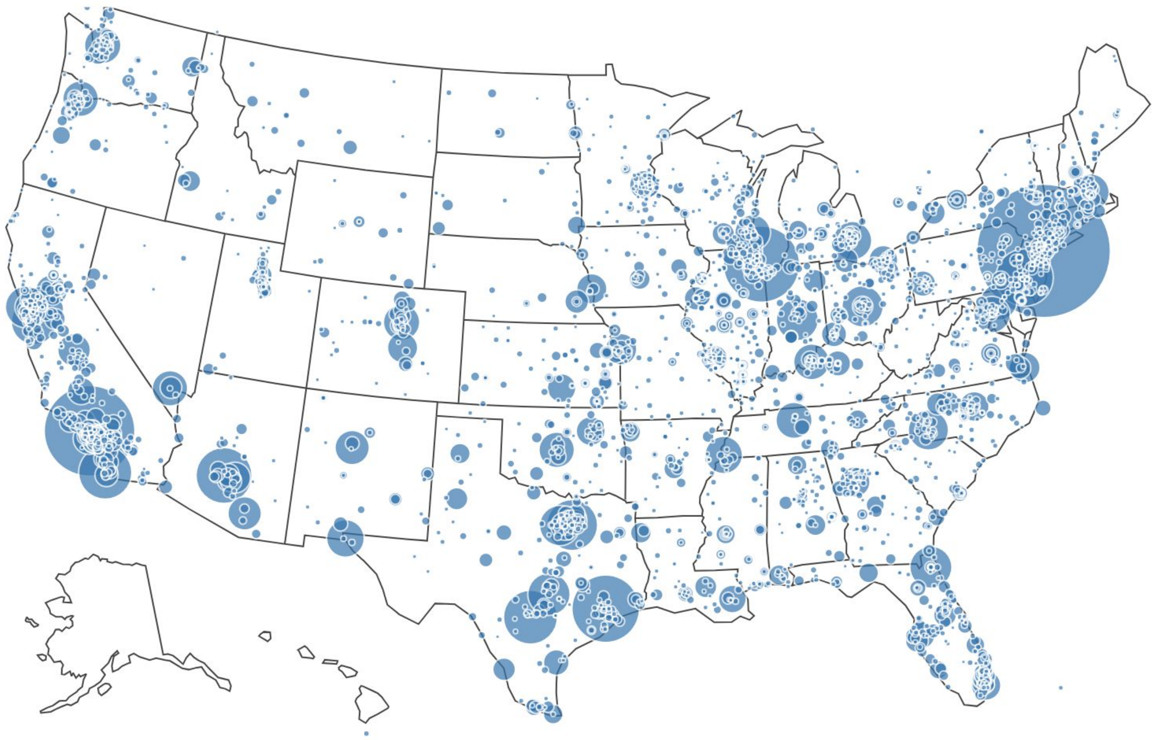

```
fig = df.plot.scatter(x='day_of_year', y='births')
```

Bubble Plots: Relationships

- Adds another variable/dimension to scatter plots, bubble/dot size
    - Can use color for even more detail
- Helps you visualize relationships between three or more numeric variables
    - Great for seeing overall patterns
    - Not ideal for precise visual comparisons as overlapping bubbles can make seeing individual points difficult



Enhancing Scatter/Bubble Charts



You can overlay onto maps for spatial visualizations.

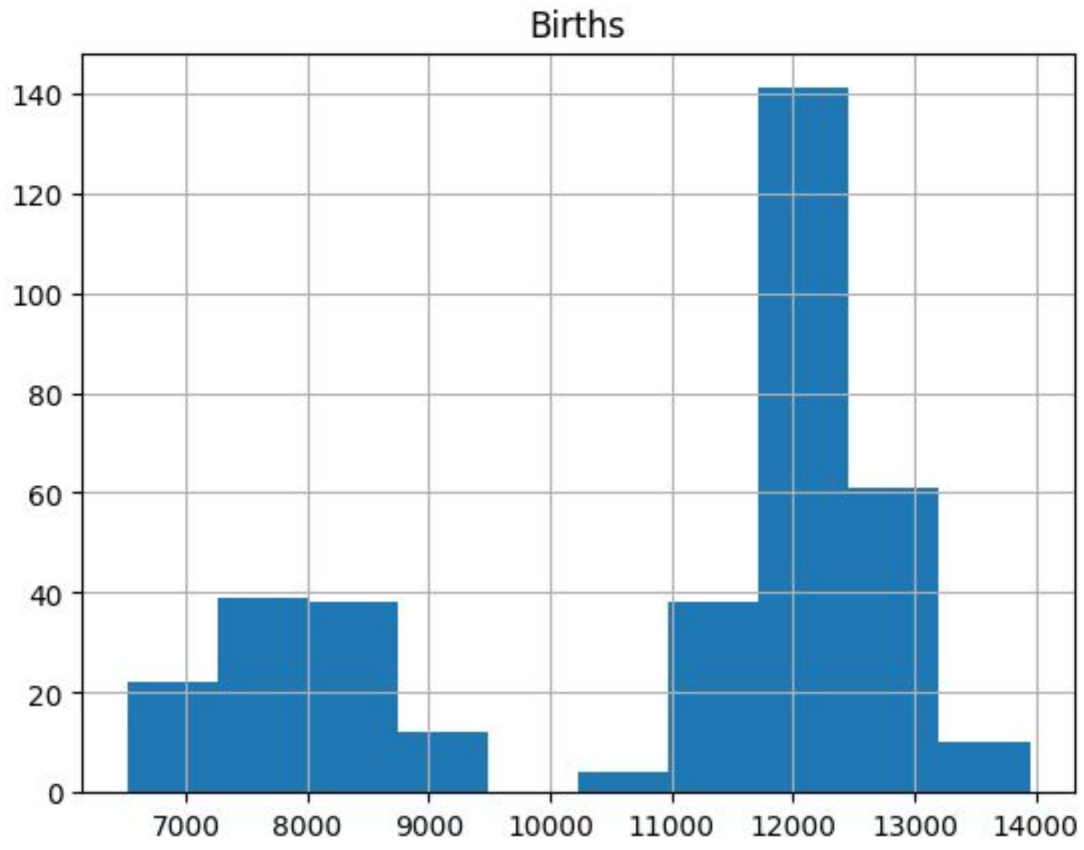
- Notice how your eye is quickly able to see geo hot spots in this example

⚠️ Careful: make sure the scale is correct.

Histogram

A histogram is a graph that shows the **frequency of numerical data,** using rectangles.

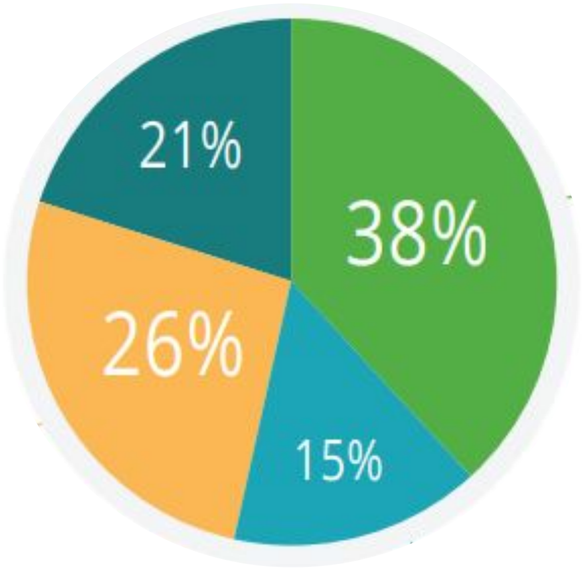
```
fig = df['births'].hist()
fig.set_title('Births')
```



Pie Charts: Clearly show proportions

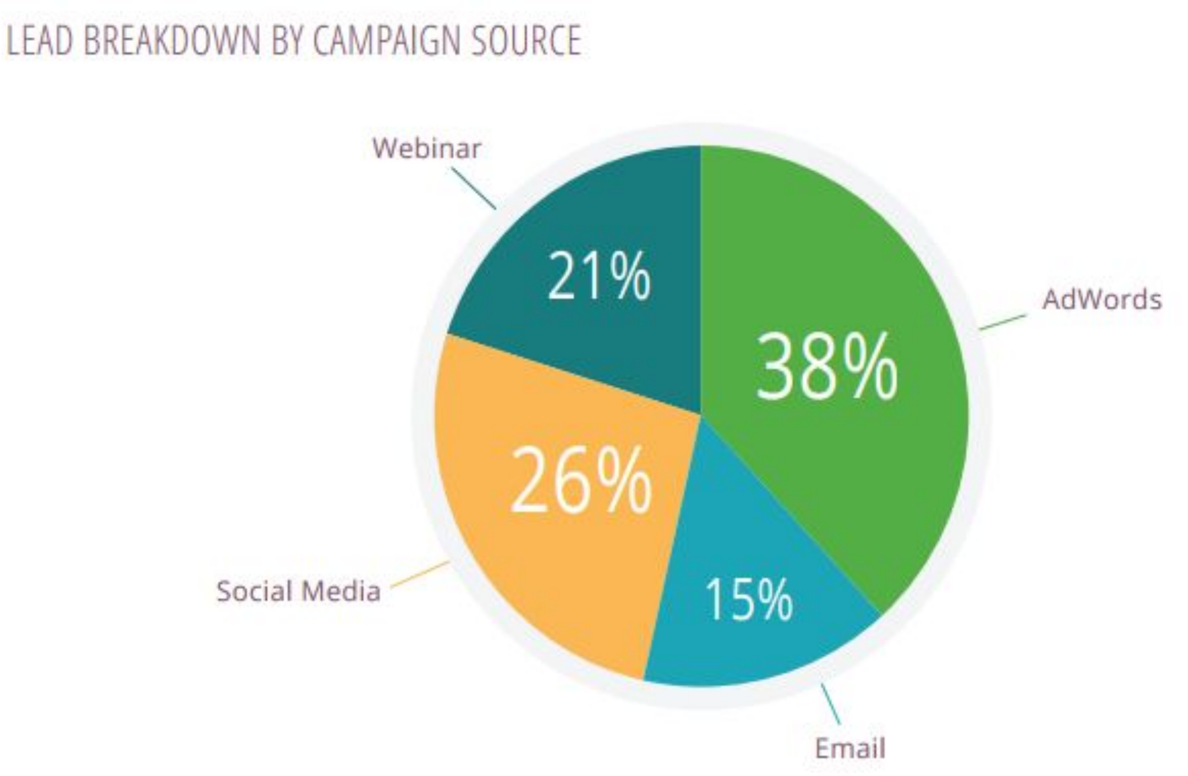
Pie charts easily show the share each value contributes to the whole.

- More intuitive than simply listing percentages that **add up to 100%**.



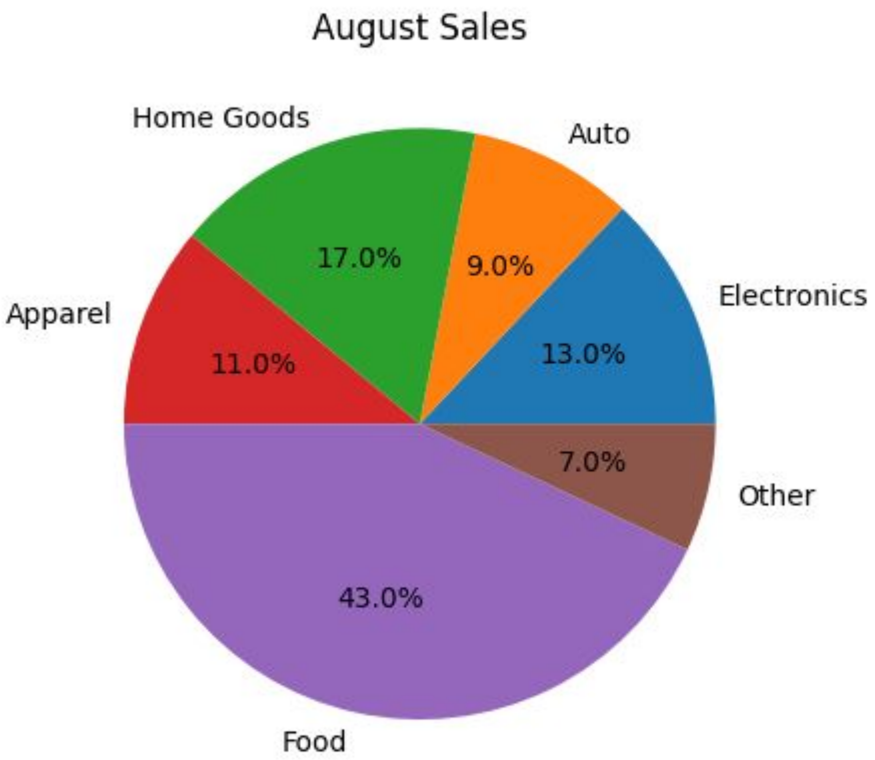
Pie Charts Example

- Clearly shows that **AdWords is the most effective**, followed by social media and webinar signups.
    - An instant insight would illuminate to the marketing team what's working best, helping them to rapidly reassign resources or refocus their efforts to maximize lead generation.



Example: This pie chart illustrates the effectiveness of different marketing campaigns in generating leads

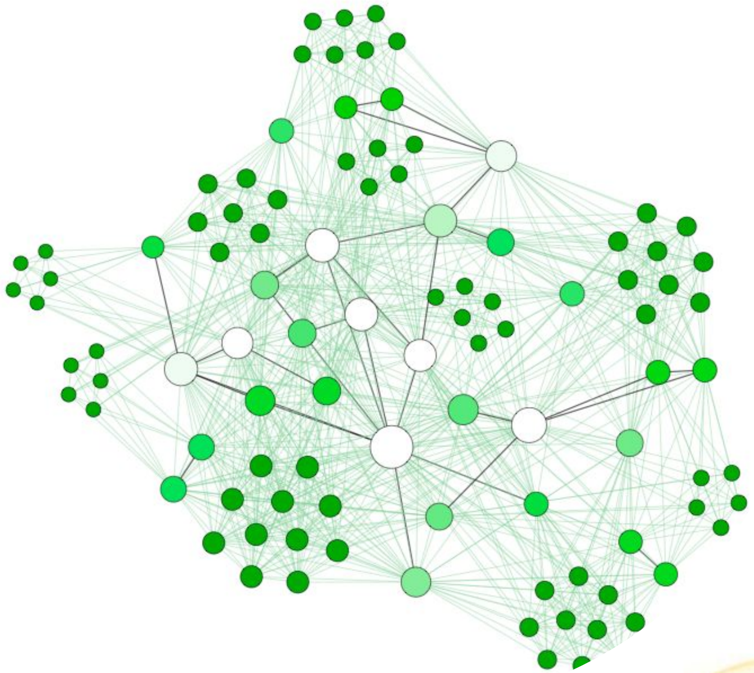
Pie Charts Example



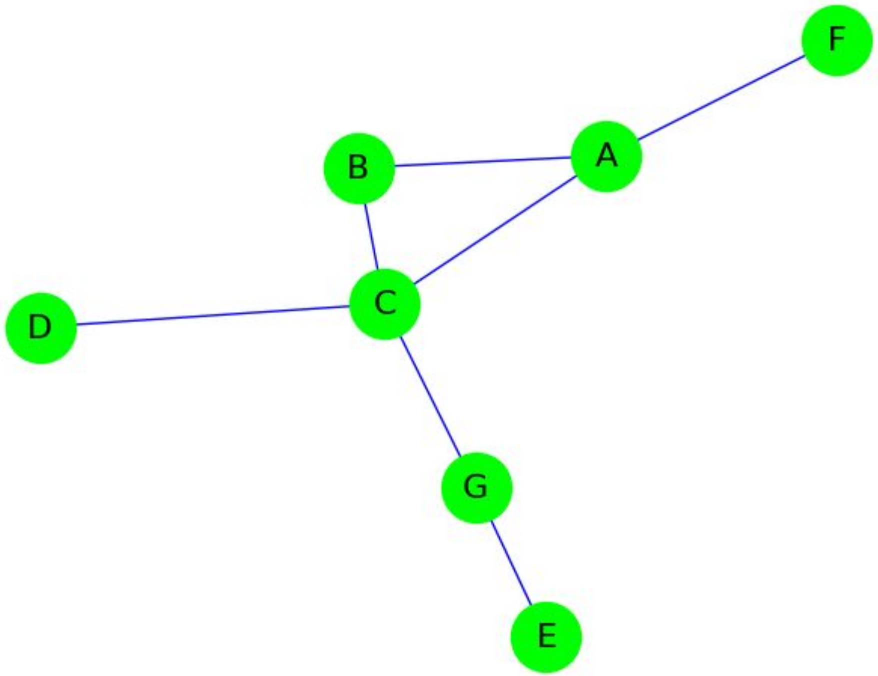

```
plt.pie(dist, labels=labels, autopct='%1.1f%%')
plt.title('August Sales')
plt.show()
```

Network Diagram

- a.k.a. Graph
- Nodes represent discrete items
  - Ex. cities
- Edges represent how they are related
  - Ex. highways, where length is distance between them
- Two nodes are “adjacent” if they are connected by a single edge
- The collection of nodes that are adjacent to a node v are called v’s “neighbors”.



Network Diagram



Simple example:

```python
import networkx as nx
import matplotlib.pyplot as plt

chart = nx.Graph()
chart.add_nodes_from(["A", "B", "C", "D", "E", "F", "G"])
chart.add_edges_from(
    [("A", "B"), ("A", "F"), ("A", "C"),
     ("B", "C"), ("C", "D"), ("C", "G"),
     ("G", "E")])

nx.draw(chart, node_color='lime', node_size=1100,
        with_labels=True, font_size=16, edge_color='blue')
plt.show()
```

Visualization using Programming

Most popular:

Python

- matplotlib
- seaborn
- plotly

R

- plotly
- ggplot2
- lattice

Common Tools for Data Visualization

- Tableau
- Microsoft Power BI
- Infogram
- ChartBlocks
- D3.js
- Google Charts
- Fusion Charts
- Chart.js

Focusing Attention

Visual Perception

Visual processing happens in its own part of the brain, based on its own rules!

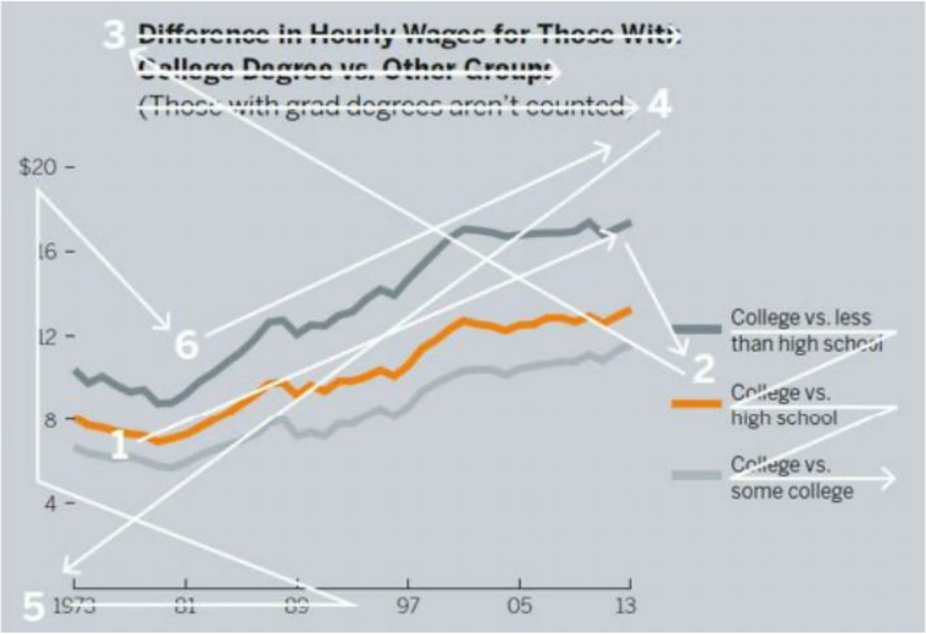
Presenters can leverage research helping us understand these rules.

- Visual Attention Theory
- Visual Salience
- Preattentive features

Visual Attention Theory

In writing, attention goes in a fixed order.

With visualizations, our eyes follow the visuals!



And you will read this last

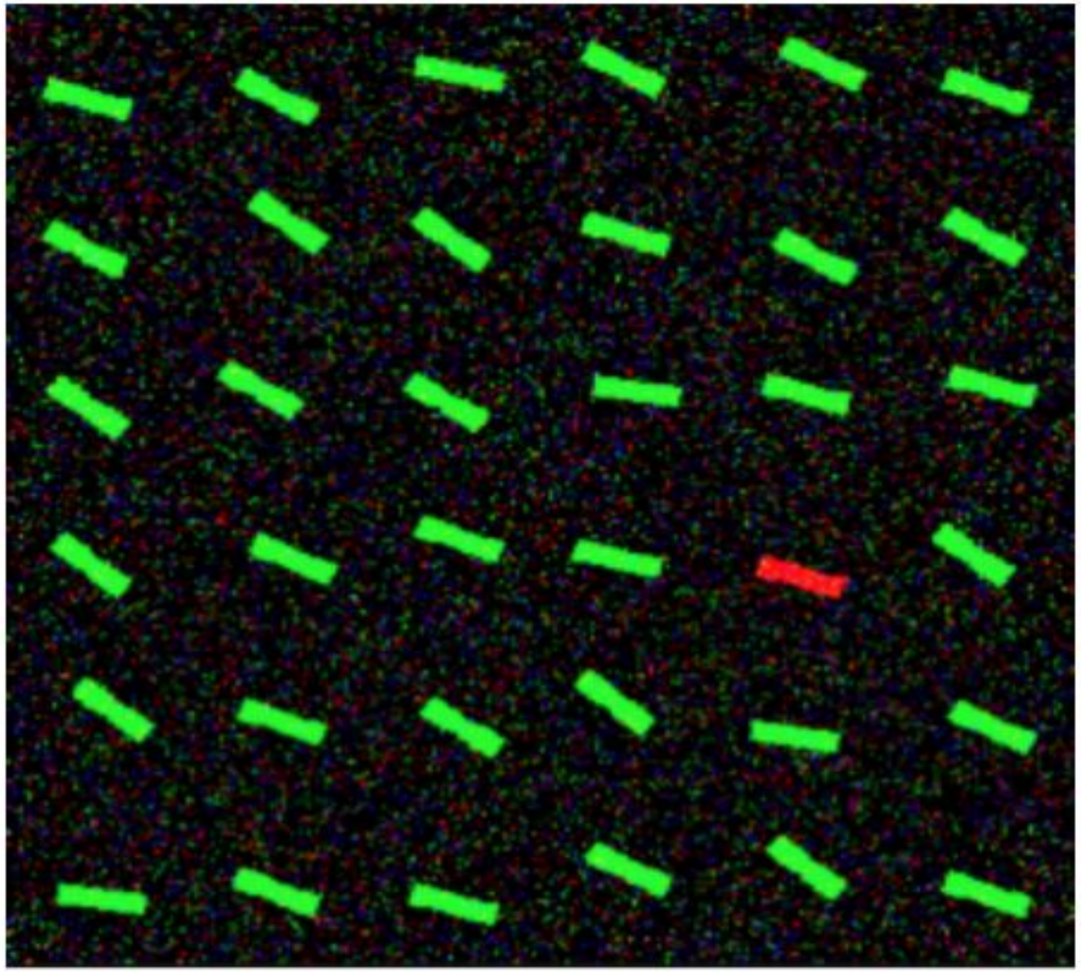
You will read this first

And then you will read this

Then this one

Visual Attention Theory

Track where your attention goes!

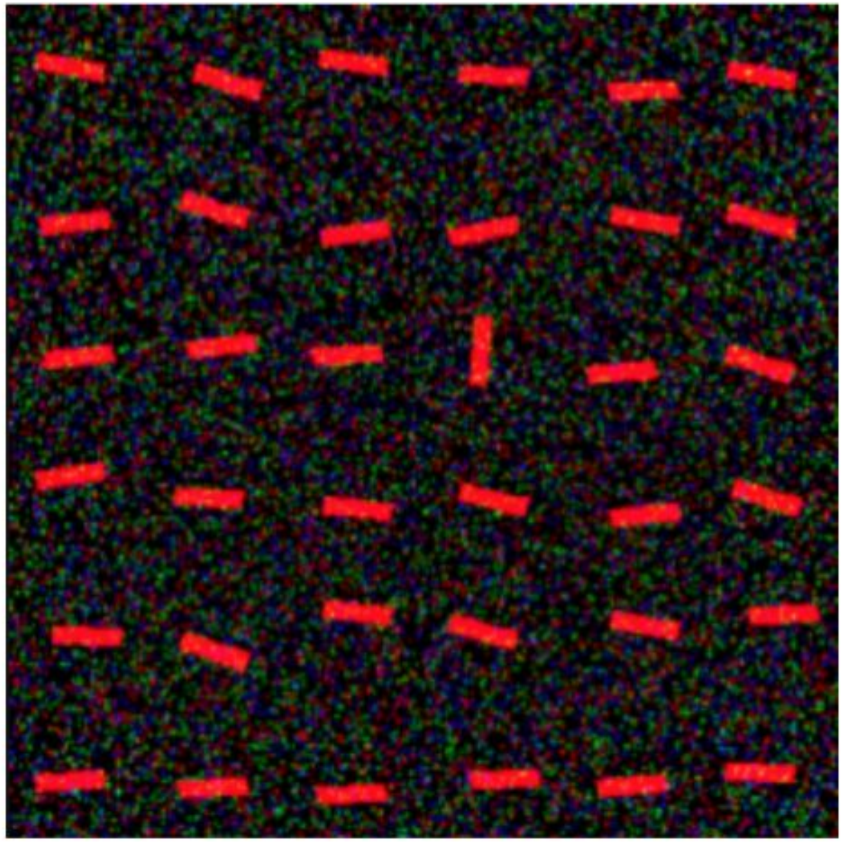


Visual Attention Theory

The red bar was salient in the first image ...but visual properties alone do not determine salience; context is crucial.

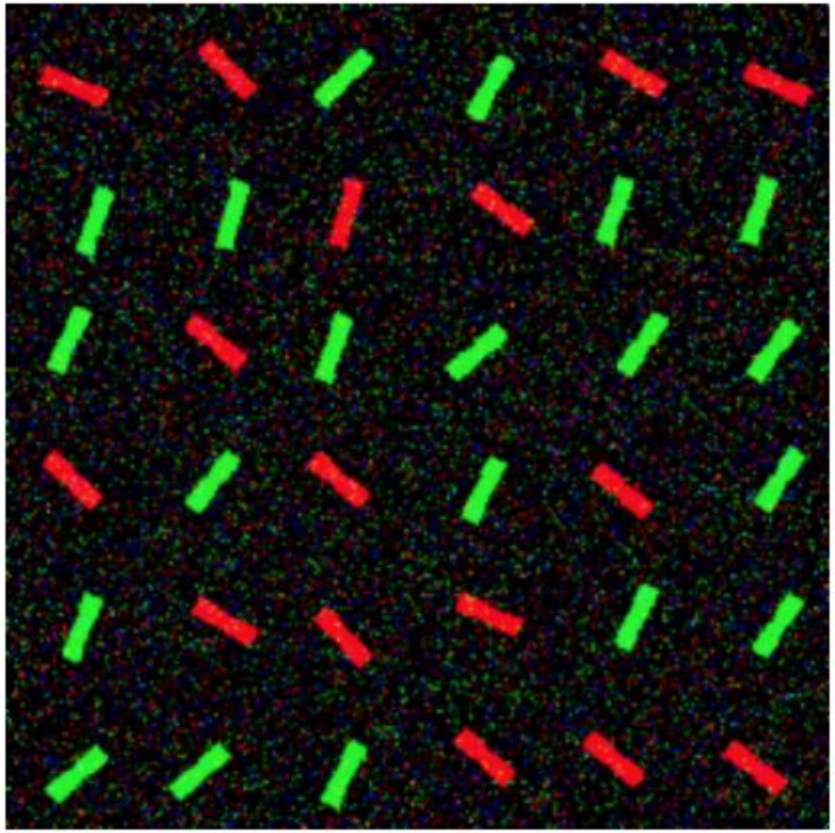
Orientation and color are examples of **preattentive features**.

Preattentive processing is performed automatically and instantly* on the entire visual field, but only for a limited set of object features.



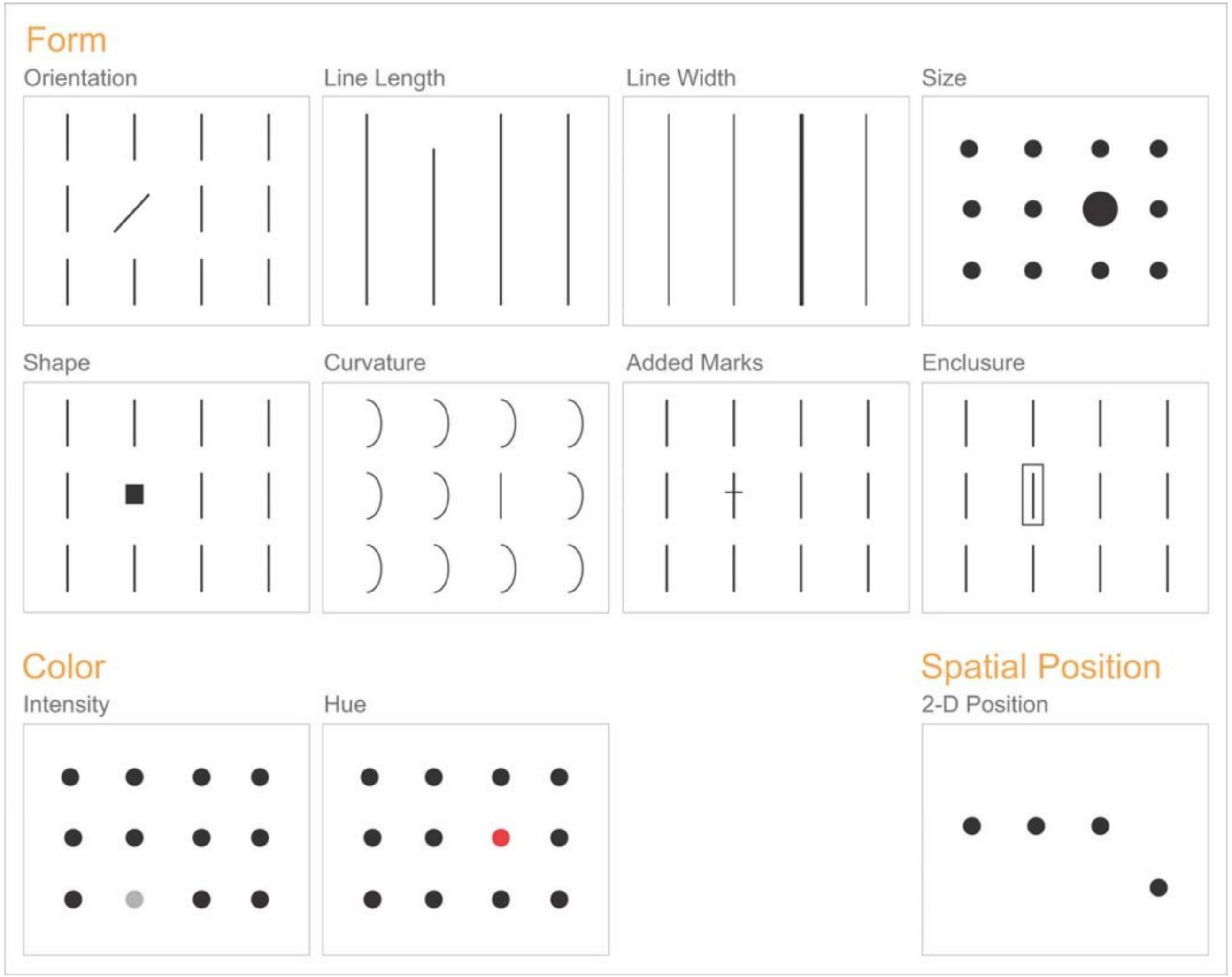
Visual Attention Theory

Harder to parse both at once!



Preattentive Features

Basic features such as color, size, or tilt (known as channels) are extracted from the display effortlessly — without the need to focus attention.



Preattentive attributes that are of particular use in visual displays of data

Using Color

Be thoughtful about how you use color.

- **If you use multiple colors, make sure they mean something!**
  - People will look for meaning in your color choices
- **Don’t abuse color**
  - Ex. making one color bright neon yellow while the others are muted shades will overly accentuate that variable
- **Don’t buck convention**
  - Ex. Using red=good and green=bad
- **Check for accessibility**
  - Ex. There are lots of available color palettes that have been thoroughly tested for being effective with both colorblind and non-coloblind people
- **Never choose style over substance**

Design Principles

- **Each visualization should be <u>designed</u> to communicate a specific insight or two**.
  - Only include graphs if they have a specific purpose
  - Know what each graph is telling your viewers
- **Maximize information, minimize ink**
  - Avoid extraneous information
- **Organize hierarchically**
  - Start broad and zoom in
- **Dazzle**
  - A good visualization is worth a lot!

References for Later

Comparison of Data Viz Libraries

- Nice Jupyter notebook comparing graphs across the 3 most popular Python data visualization libraries, matplotlib, seaborn, and plotly (uses express)
    - https://nbviewer.org/github/dylanjcastillo/random/blob/main/cheatsheet_data_viz_python.ipynb

Matplotlib

- Cheat Sheets
  - https://matplotlib.org/cheatsheets/_images/cheatsheets-1.png
  - https://matplotlib.org/cheatsheets/_images/cheatsheets-2.png
- Handouts
  - Beginner: https://matplotlib.org/cheatsheets/_images/handout-beginner.png
  - Intermediate: https://matplotlib.org/cheatsheets/_images/handout-intermediate.png
  - Tips: https://matplotlib.org/cheatsheets/_images/handout-tips.png

More at https://matplotlib.org/cheatsheets

Seaborn

- Python Graph Gallery quick reference
  - https://python-graph-gallery.com/seaborn
- Nice quick reference on Kaggle
  - https://www.kaggle.com/code/themlphdstudent/cheat-sheet-seaborn-charts
- DataCamp Seaborn Cheat Sheet
  - https://images.datacamp.com/image/upload/v1676302629/Marketing/Blog/Seaborn_Cheat_Sheet.pdf